

Data Analytics Capstone
Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S.
Manufacturing

Interim Report

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Code and notebooks (GitHub): <https://github.com/somndas3/qm640-tri-pollution-prevention-capstone>

Raw and processed data, view-only (Google Drive):

https://drive.google.com/drive/folders/1oPNHYCAvGwc7uQ1iIP2a_VlZdLnFJuid?usp=sharing

Original data source (U.S. EPA): <https://www.epa.gov/toxics-release-inventory-tri-program/tri-basic-plus-data-files-calendar-years-1987-present>

Pollution-Prevention Actions and Next-Year Toxic Release Reduction in U.S.

Manufacturing

Background and Context

Every year, thousands of U.S. manufacturing plants report how much of each toxic chemical they released and what they did to release less. The U.S. Environmental Protection Agency (EPA) collects these filings through the Toxics Release Inventory (TRI), a public program whose data allow American industry's environmental behavior to be studied openly and reproducibly (U.S. Environmental Protection Agency [EPA], 2025b, 2025c, 2025d).

Underneath the program sits a clear policy preference: the Pollution Prevention Act of 1990 holds that preventing waste at the source is better than recycling, treating, or disposing of it (EPA, 2025a). Source reduction changes how a plant works so the waste is never created, and facilities report such changes using standard activity codes, so in principle a plant reporting one is a plant actively trying to release less.

That creates a practical question. Plant managers, environmental health and safety (EHS) leaders, and process engineers want to know whether reported actions line up with lower next-

year releases, and whether some kinds do so more strongly. A simple before-and-after comparison cannot answer this, because releases also fall for reasons unrelated to prevention.

Earlier TRI research has mostly asked larger questions: whether disclosure alone changes behavior, whether markets punish polluters, and whether prevention reduces releases over long periods (Gamper-Rabindran, 2006; Hamilton, 1995; Konar & Cohen, 1997; Ranson et al., 2015), or whether cleaner production pays for itself (Cagno et al., 2005; King & Lenox, 2002). This project is narrower: it uses recent data only, defines one outcome a plant manager can understand, and asks how well reported actions explain and predict it—a screening study, not a causal one.

Problem Statement

U.S. manufacturing facilities report many pollution-prevention actions each year, but EHS and plant leaders have no simple, recent-data framework for judging whether records carrying those actions are more likely to achieve a meaningful reduction in the same chemical's on-site release the following year. This project addresses that gap using manufacturing Form R records from 2020 through 2024, defining a meaningful reduction as a next-year release of no more than 90% of the current-year release. The result is a transparent way to prioritize review effort, not a replacement for engineering judgment and not a claim of causality.

Purpose of the Study

The purpose is to explain and predict meaningful next-year toxic-release reductions at the facility-chemical-year level using EPA TRI Basic Plus data. It describes recent patterns, tests whether reporting any action is associated with a next-year reduction after adjustment, compares each action type against the no-action group, and evaluates an interpretable model on a future-

year holdout kept untouched until the end—flagging, in business terms, the records least likely to improve next year.

Interim Project Status

This report is submitted at the end of Week 4, whose objectives were to collect the data with minimal redundancy and adequate accuracy and to prepare it for the solution approach.

- **Completed.** All ten source files were collected and checksum-verified, cleaned through a fully reconciled audit trail, and matched into four consecutive-year transitions, with the reserved 2023-to-2024 transition stripped of every outcome-bearing column. Exploratory analysis and the justified choice of method for each research question are also complete.
- **In progress.** Feature-engineering decisions from the exploratory findings, notably the production-ratio functional form, and the code for the adjusted RQ2 association model with facility-clustered standard errors.
- **Pending.** Fitting the RQ2 adjusted logistic regression, running both sensitivity analyses, building and locking the RQ4 elastic-net screening model against its baseline, and the single reserved-holdout evaluation.

Remaining work is scheduled by week under Next Steps.

Scope and Objectives

Population and Outcome

The population is U.S. TRI-reporting manufacturing facilities whose primary North American Industry Classification System (NAICS) code begins with 31, 32, or 33. The unit of analysis is one facility, one chemical, one reporting year. Only Form R records are retained, because the shorter Form A carries no release quantity.

The primary outcome, REDUCTION_10, equals 1 when the next-year on-site release is no more than 90% of the current-year release and 0 otherwise. It is defined only when the current-year release exceeds zero, because a percentage change needs a non-zero denominator. A record absent the following year is excluded rather than scored as a reduction. Dioxins are reported in grams rather than pounds and are never mixed with pound-reported chemicals in any quantity-based calculation.

Research Questions

RQ1: Do meaningful next-year reduction rates differ across action years and major manufacturing subsectors?

RQ2: After controlling for current release level, production or activity ratio, manufacturing subsector, chemical hazard classification, and action year, is reporting at least one pollution-prevention action associated with achieving a meaningful next-year reduction?

RQ3: Do next-year reduction rates differ between each reported pollution-prevention action type and records with no reported action?

RQ4: Can an interpretable model using only action-year information predict which facility-chemical records will fail to achieve a meaningful reduction in the following year better than chance?

Literature Survey

Literature Review Approach

Sources fall into two groups. Subject-matter sources on the TRI program, public environmental disclosure, and source reduction in manufacturing were found through Google Scholar and EPA documentation and retained when they reported an empirical result for U.S. facility-level data. Method sources were selected because each governs one decision in this

study: sizing a logistic sample, handling repeated observations from the same facility, correcting for multiple comparisons, and judging a prediction model honestly. [Table E1](#) in Appendix E maps every source to the research question or design decision it supports, giving each article's domain and context, method, key finding, and relevance to this project.

Themes in the Literature

Public reporting changes behavior, but not always as the raw numbers suggest. Hamilton (1995) found the first TRI disclosure moved share prices, and Konar and Cohen (1997) found the firms punished hardest cut emissions most; yet Gamper-Rabindran (2006) showed part of the apparent national decline reflected reporting changes rather than real abatement. Hence the outcome here is deliberately narrow—same facility, chemical, and unit, one year later—across RQ1–RQ4.

Prevention is not one thing. King and Lenox (2002) located the profitable part of pollution reduction in preventing waste rather than treating it; Cagno et al. (2005) found payback varied widely across 134 projects, with equipment and process changes best; and Ranson et al. (2015), the closest prior study, linked source-reduction activity to lower reported releases. These justify comparing the five action types separately in RQ3 rather than pooling them.

Methodological discipline maps onto the design. Because one facility contributes many records, observations are not independent, and Cameron and Miller (2015) show ignoring this understates standard errors—hence facility-clustered errors in RQ2. Because five action types are tested at once, Holm's (1979) correction applies in RQ3. Because the RQ4 model must support a decision, Steyerberg et al. (2010) and Van Calster et al. (2019) require calibration alongside discrimination, and Peduzzi et al. (1996), Riley et al. (2019), and van Smeden et al. (2019) supply three sample-size criteria, of which the study adopts the most demanding.

Data Description

Data Sources and Access

The data come from the EPA TRI Basic Plus Data Files (EPA, 2025c), with field definitions from the accompanying guides (EPA, 2025d). Ten files are used—File Type 1A and 2A for each year 2020 through 2024—joined within each year on the document control number. File 1A supplies facility and chemical identifiers, industry code, unit of measure, hazard flags, and total on-site releases; File 2A supplies the production ratio and up to four source-reduction activity codes. The fixed-layout parsing rule is recorded in Appendix B.

Dataset Overview

The characteristics of the analysis dataset are as follows.

- **Records.** 315,581 raw Form 1A rows across action years 2020–2023, reducing to 163,501 eligible matched records. Reporting-year 2024 File 1A is used to establish the reserved next-year outcome, which is withheld from EDA and modelling. File 2A for 2024 is used only for descriptive verification of the activity-code scheme and is not used for feature selection, model fitting, tuning, or evaluation.
- **Variables.** 23 non-outcome columns: 4 identifiers, 15 predictors, 4 audit fields, plus 3 outcome columns existing only in the development data.
- **Time period.** TRI reporting years 2020 through 2024, forming four consecutive-year transitions.
- **Unit of analysis.** One facility, one TRI chemical, one reporting year.
- **Target variable.** REDUCTION_10, equal to 1 when the next-year on-site release is at most 90% of the current-year release.

The eligible records split into 123,535 development records covering the first three transitions and 39,966 reserved records covering 2023-to-2024. The development records contain 38,890 reductions, a rate of 31.48%. The complete data dictionary, defining every analysis variable, is [Table E2](#).

Sample-size adequacy after cleaning. Cleaning removes slightly over half of the raw rows, so retained counts were checked against the planning minimums rather than assumed adequate, and every requirement is met with a wide margin ([Table E3](#)). RQ1 and RQ3 use all eligible same-unit records because their outcome is calculated within chemical and unit, while RQ2 and RQ4 use the pound-reported modelling frame because current release enters as a quantitative predictor; gram-reported dioxin records are evaluated separately in sensitivity analysis.

Data Availability

All materials needed to reproduce this analysis are openly available in view-only mode.

- **Raw data.** The ten TRI Basic Plus source files, File 1A and File 2A for reporting years 2020 through 2024, organized in one folder per year:
https://drive.google.com/drive/folders/1oPNHYCAvGwc7uQ1iIP2a_VlZdLnFJuid?usp=sharing
- **Processed data.** The same Drive location, processed subfolder: the development dataset, the features-only holdout, the data dictionary, the preprocessing parameters, and seven audit tables.
- **Code and notebooks.** The exploratory notebook that produced every figure in Appendix A, plus the preparation pipeline: <https://github.com/somndas3/qm640-tri-pollution-prevention-capstone>

- **Original source.** The files were downloaded from the EPA TRI Basic Plus page, which remains the authoritative source: <https://www.epa.gov/toxics-release-inventory-tri-program/tri-basic-plus-data-files-calendar-years-1987-present>

The ten raw files total 535.6 MB, exceeding practical repository limits, so they are distributed through Drive rather than GitHub. [Table B1](#) in Appendix B lists each file's size and SHA-256 checksum, and [Table B2](#) records the consolidation counts by reporting year; Figures D1 through D4 in Appendix D evidence the folder structure, the processed dataset contents, a preview of the development data, and the activity-code crosswalk.

Analysis

Data Cleaning

Cleaning was treated as an auditable process. Every filter writes a counted row into a reconciliation log required to close exactly against the raw counts. [Table E4](#) records each issue and its treatment, and [Table E5](#) gives the full record-count waterfall from raw files to analysis sets.

Two items were resolved differently from the original plan. For duplicates, where any filing in a facility-chemical-year group carries a revision code the revision supersedes; otherwise releases are summed using a minimum-count setting, which across five years meant summing in 1,551 groups and supersession in 127. Appendix C gives the exact deterministic rule.

The second is the activity-code scheme change. In 2020 the EPA used legacy W codes; from 2021, the S scheme. Because the study spans that boundary, the same action appears under two labels depending on year. All 70 observed codes were mapped onto the five action types, and the pipeline asserts none is unmapped ([Figure A1](#)). The consequence is that a rise in reported actions between 2020 and 2021 cannot be read as a pure change in behavior.

The waterfall runs from 315,581 raw rows to 163,501 eligible records, splitting into 123,535 development and 39,966 reserved. The eligibility filter removes 41,946 records—more than twice the 18,738 lost to matching—mostly base-year releases recorded as zero, so the analysis is conditioned on records already releasing something and says nothing about facilities already at zero. Duplicate consolidation removes only 2,176 records, under 1% of the manufacturing total.

Exploratory Data Analysis

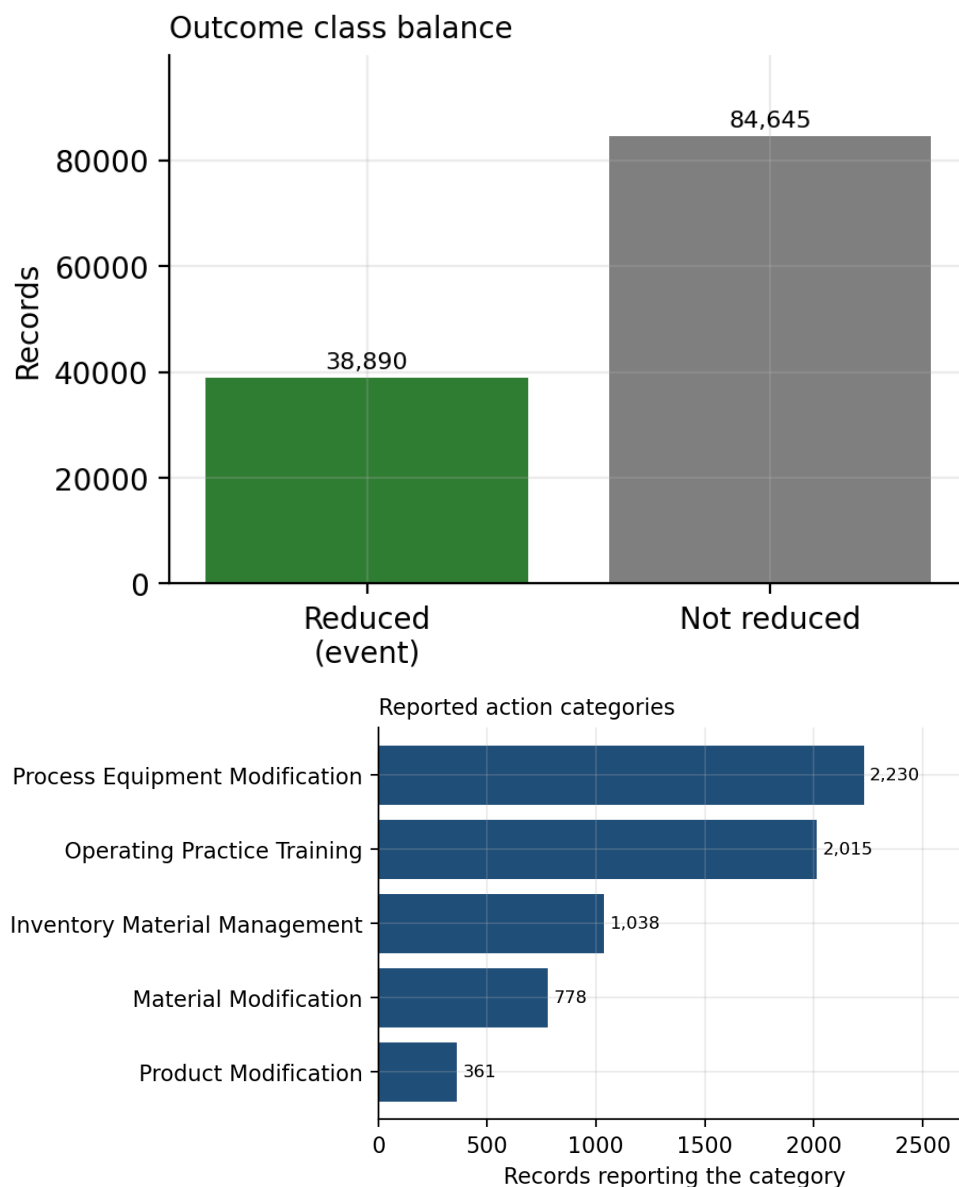
The exploratory work had three jobs: describe the data honestly, test the assumptions the models rely on, and surface anything that would change the plan. The exploratory figures appear in the main text and in Appendix A, each referenced where it is discussed, and every confidence interval is a Wilson score interval (Agresti & Coull, 1998). [Table E6](#) summarizes each figure, its insight, and the decision it drove.

Outcome Is Balanced, but Reported Action Is Rare

The outcome is well balanced: of 123,535 development records, 38,890 achieve a reduction of at least 10%, a rate of 31.48% ([Figure 1](#)). No resampling is needed. The exposure, however, is rare. Only 6,012 records, or 4.87%, report any pollution-prevention action, and that rarity is the central statistical constraint of the project.

Figure 1

Outcome Class Balance and Reported Action Categories



Note. Of 123,535 development records, 31.48% achieve a reduction of at least 10%. Only 4.87% report any pollution-prevention action.

Release quantities are extremely skewed ([Figure A2](#)). Across 122,017 pound-reported records the mean current-year release is 24,737 lb against a median of only 217 lb, with a maximum of 19,750,398 lb and skewness of 34.6. These extremes were inspected and kept as genuine large emitters, but they mean release cannot enter a model on its raw scale. A log transformation is used instead, and [Figure A10](#) shows the empirical logit generally increases with

log current release, supporting the log transformation, though some departures from linearity remain and will be checked against a flexible alternative during development-model diagnostics.

The production or activity ratio is centered near 1.00, indicating that most facilities reported production similar to the prior year ([Figure A4](#)). The field is missing for 1.62% of development records ([Figure A5](#)). Missingness is low, but its mechanism cannot be established from observed data. For modelling, missing values will be replaced with the median calculated from the applicable training fold, and a separate missingness indicator will be retained. The median will never be calculated using validation or holdout records. A complete-case sensitivity analysis will be used to assess whether this treatment materially affects the results.

The ratio also holds extreme but valid values that are implausible as literal production multipliers. It will be capped at the training-fold 99th percentile before centering and squaring, with the original retained in an audit field; the imputation median, caps, and centering value are applied unchanged to validation data and confirmed by development-only sensitivity testing.

Decision. Keep the full sample without resampling, and treat rare action reporting—not class imbalance—as the central statistical constraint. Release enters models on the log scale, and the production ratio is imputed, capped, and centered inside each training fold.

A Coding-Scheme Change Inflates 2021 Action Counts

[Figure A1](#) shows legacy W codes dominating 2020 then replaced by S codes; all 70 distinct codes map onto the five action types with zero unmapped, which the pipeline asserts. [Figure A3](#) shows why this matters: process or equipment modification rises from 1.32% in 2020 to 2.13% in 2021 and material modification from 0.41% to 0.87% across the same boundary. Because the jump coincides exactly with the scheme change, it may partly reflect the coding change rather than behavior; the other three types stay flat, suggesting a specific effect.

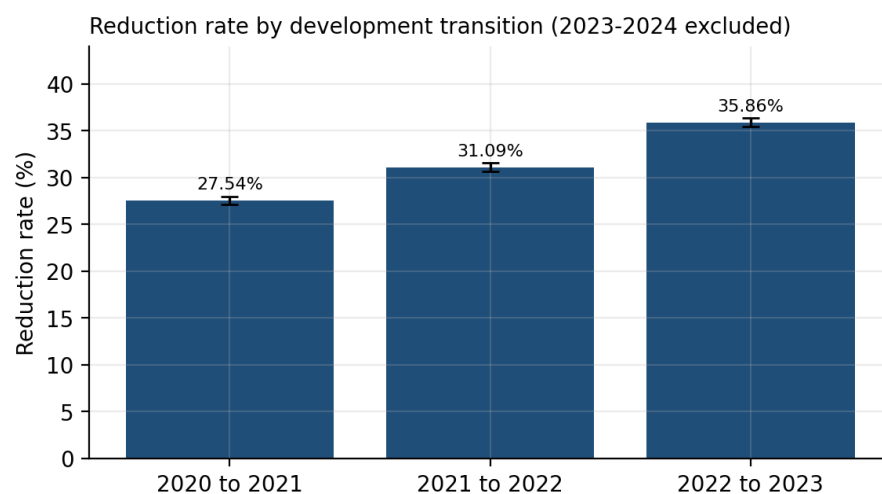
Decision. Retain the mapped five-type action scheme, but never read the 2020-to-2021 rise in reported actions as purely behavioral; the sensitivity analysis excluding that transition is mandatory.

Reduction Rates Rise Over Time and Vary by Industry

Reduction rates rise steadily across the three development transitions ([Figure 2](#)): 27.54% [27.11, 27.97], then 31.09% [30.65, 31.54], then 35.86% [35.39, 36.32]. No pair of intervals overlaps, and $\chi^2(2) = 663.72$, $p < .001$. But the effect size is the useful part: Cramér's V is .073, a weak association. Year matters reliably yet modestly. The pattern may partly reflect recovery from the 2020 pandemic disruption, although other temporal and reporting changes may also contribute. Year is kept as a control, and the key comparisons will be repeated without the 2020-to-2021 transition.

Figure 2

Reduction Rate by Development Transition



Note. Rates rise from 27.54% to 31.09% to 35.86% with non-overlapping 95% intervals. The reserved 2023-to-2024 transition is excluded by design.

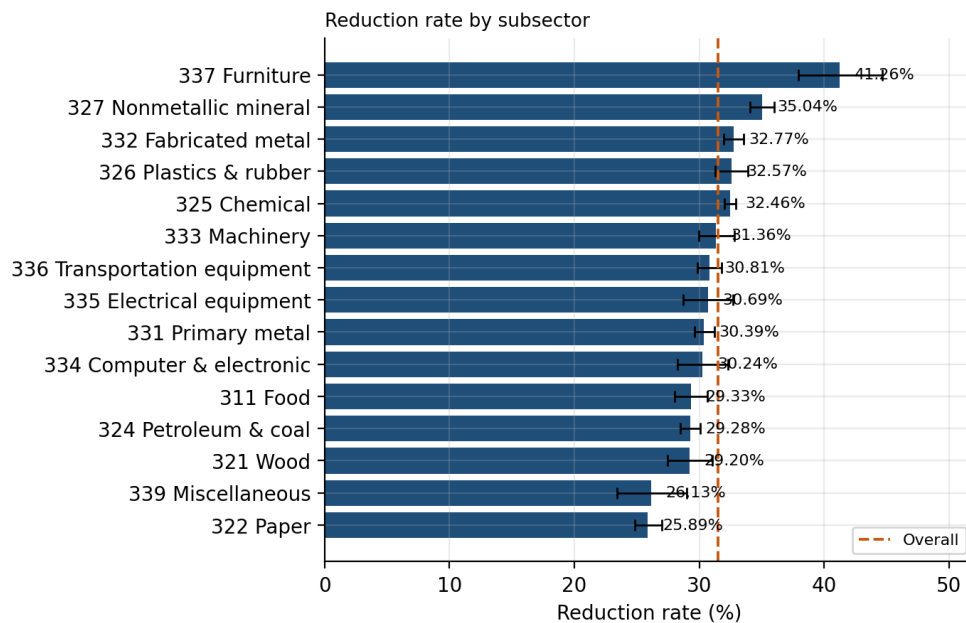
Industry matters more visibly. [Figure 3](#) shows all 15 qualifying subsectors, from 25.89% in paper (NAICS 322) to 41.26% in furniture (NAICS 337), a spread of 15.4 points; $\chi^2(14) =$

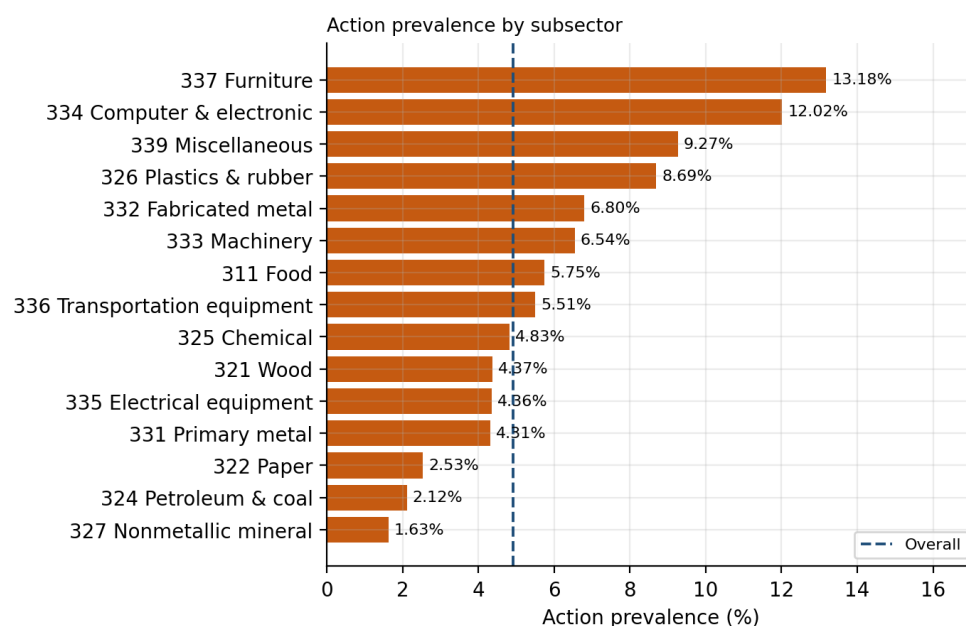
280.04, $p < .001$, Cramér's $V = .048$. The second panel carries a warning that shapes the modeling: the subsectors reporting most actions are not those reducing most often. Nonmetallic mineral products report at only 1.63% yet achieve the second-highest rate of 35.04%, while computer and electronic products report at 12.02% and sit mid-table at 30.24%. Any unadjusted action comparison is therefore partly a comparison of industries.

The 15 subsectors shown in [Figure 3](#) meet the prespecified minimum-count requirement. Records from smaller manufacturing subsectors will not be treated as Chemical Manufacturing, which is the reference category. They will be combined into an explicit Other Manufacturing category for modelling. This preserves the full manufacturing population while preventing small groups from producing unstable individual coefficients.

Figure 3

Reduction Rate and Action Prevalence by Manufacturing Subsector





Note. All 15 qualifying subsectors are shown. Rates span 15.4 points, and the subsectors reporting the most actions are not those reducing most often.

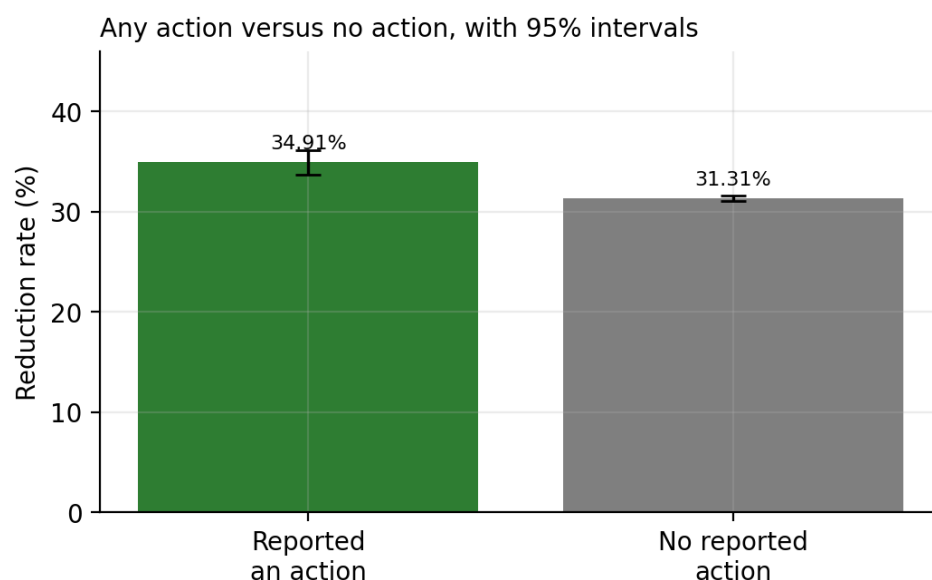
Decision. Keep action year and subsector as controls, collapse below-threshold subsectors into an explicit Other Manufacturing category, and repeat key comparisons without the 2020-to-2021 transition.

The Unadjusted Action Gap Is Small but Positive

Records reporting any action reduce at 34.91% ($n = 6,012$) against 31.31% for those reporting none ($n = 117,523$), a gap of 3.61 points, 95% CI [2.37, 4.84], relative risk 1.12, $p < .001$ ([Figure 4](#)). It would be easy to conclude prevention works, but two findings block that. First, the groups are not comparable: seven of 19 covariates differ beyond the conventional 0.10 standardized difference ([Figure A9](#)), the largest absolute standardized mean differences are 0.244 and 0.232, both reflecting differences in industry composition, then persistent, bioaccumulative, and toxic (PBT) status at 0.15 and action year 2020 at 0.13. Second, there is a mechanical gradient in the outcome itself, described next. [Figure A11](#) shows the association holds in direction across strata, including without the pandemic-affected transition, but direction is not magnitude and none of it is adjusted.

Figure 4

Reduction Rate for Any Reported Action Versus No Reported Action



Note. The unadjusted gap is 3.61 percentage points, 95% CI [2.37, 4.84]. This comparison is not adjusted for industry, release level, or year.

Decision. The raw 3.61-point gap cannot be interpreted until industry, release level, and year are adjusted; this is precisely the RQ2 question.

Release Level and Production Ratio Confound the Comparison

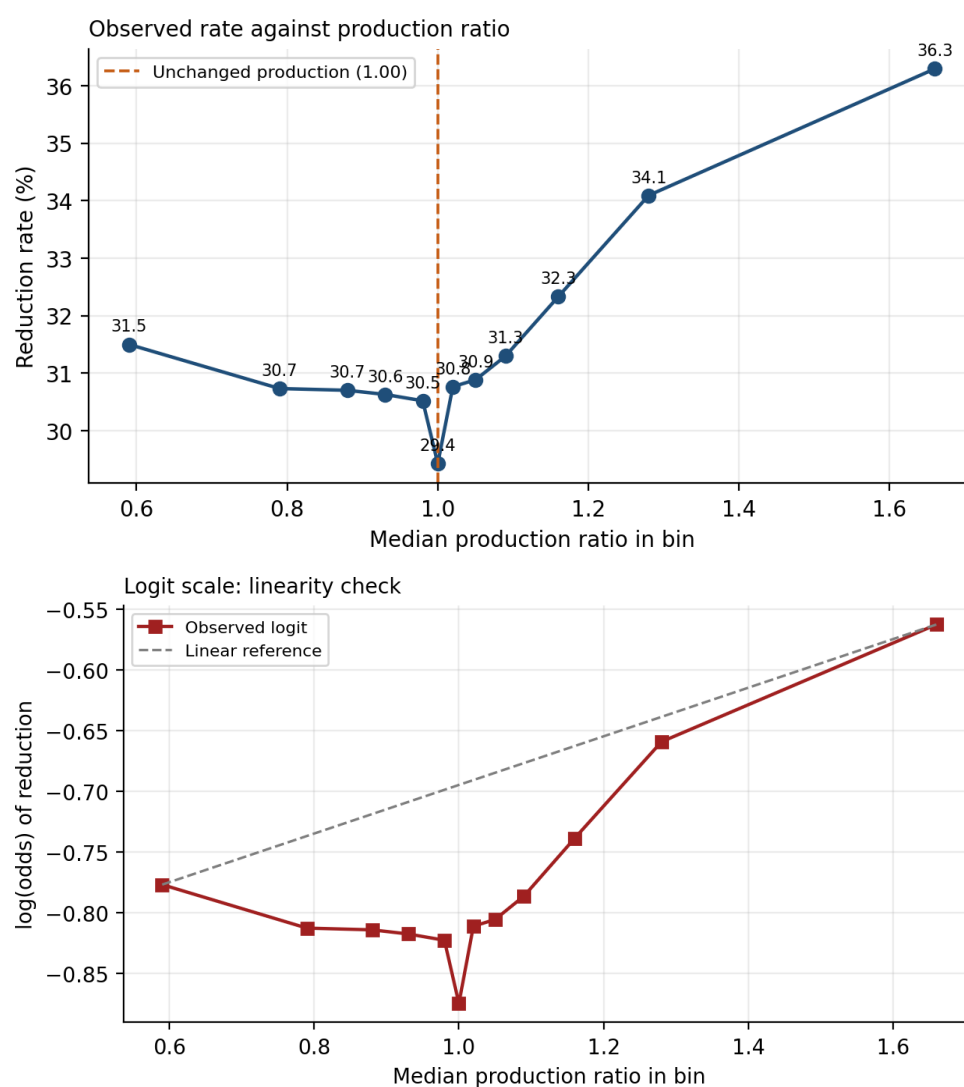
Current release relates to meeting the 10% reduction threshold, though not monotonically: [Figure A6](#) shows lower rates in some low-release deciles and higher rates in several upper deciles, consistent with the starting level shaping the opportunity for a percentage reduction and possibly regression to the mean. Current release is therefore retained as a mandatory adjustment variable.

The production ratio behaves non-monotonically ([Figure 5](#)): reduction rates are highest at both ends and lowest near a ratio of 1.00, consistent with falling production cutting releases directly and sharply rising production accompanying new equipment or process changes, though neither mechanism is demonstrated. Because the relationship is curved, an Akaike information

criterion comparison of three specifications gives the quadratic a modest edge over the linear form and a decisive one over the logarithmic (Akaike, 1974; Burnham & Anderson, 2002); it is carried as provisional, to be tested inside the full adjusted model with the ratio centered before squaring to prevent artificial collinearity (Aiken & West, 1991). Tables E4 and E7 document the resulting preprocessing and feature decisions.

Figure 5

Observed Reduction Rate Against the Production Ratio, with Logit Linearity Check



Note. The relationship is U-shaped, with a minimum of 29.72% near a ratio of 1.00 and 36.19% in the top decile. A centered quadratic term is carried as provisional.

Decision. Force current release into every model and carry a centered quadratic production-ratio term, to be confirmed inside the full adjusted model rather than adopted on a single-variable comparison.

Two of Five Action Types Stand Out

RQ3 compares each action type against the no-action group separately. [Table 1](#) reports the results with Holm-adjusted p-values and [Figure 6](#) shows the effect sizes with their intervals.

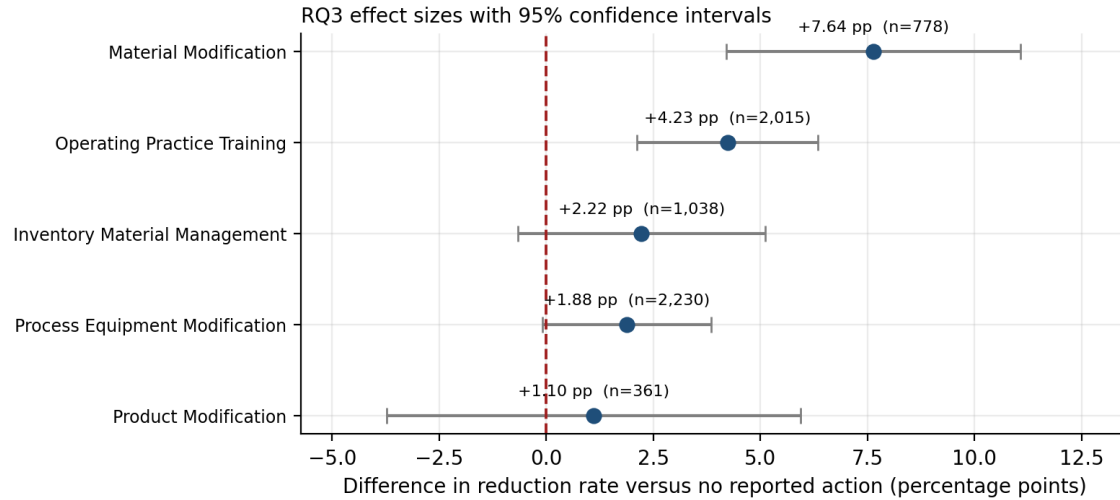
Two of the five comparisons survive the correction: material modification at 7.64 percentage points, relative risk 1.24 [1.14, 1.36], and operating practice or training at 4.23 points, relative risk 1.14 [1.07, 1.21]. The other three cannot be distinguished from the reference group. Process or equipment modification is the near miss, with a relative-risk lower bound of 0.999 just touching 1.00 and a Holm-adjusted p of .175.

The three nonsignificant comparisons do not support differences as large as the prespecified 10-percentage-point planning effect. Their confidence intervals exclude effects of that magnitude but remain compatible with smaller positive, null, or slightly negative differences. The correct interpretation is therefore that no difference of the planned magnitude was detected, not that the action type has no association.

All effect sizes are tiny, with a maximum Cramér's V of .0133. That is the tension in this project: the differences are statistically clear because the sample is enormous, yet small and unadjusted. Whether they survive the controls is the RQ2 question.

Figure 6

Rate Differences by Action Type with 95% Confidence Intervals



Note. Material modification and operating practice or training have 95% confidence intervals that exclude zero and remain statistically significant after Holm adjustment. The other three comparisons do not remain statistically significant after adjustment.

Table 1

Reduction Rate by Action Type Compared With Records Reporting No Action

Action type	n	Rate (%)	95% CI	Difference (pp)	95% CI of difference	Relative risk	95% CI of RR	Holm <i>p</i>
Material modification	778	38.95	[35.58, 42.42]	7.64	[4.20, 11.08]	1.244	[1.139, 1.359]	< .001
Operating practice or training	2,015	35.53	[33.47, 37.65]	4.23	[2.12, 6.33]	1.135	[1.070, 1.205]	.0002
Inventory or material management	1,038	33.53	[30.72, 36.45]	2.22	[-0.66, 5.10]	1.071	[0.983, 1.167]	.249
Process or equipment modification	2,230	33.18	[31.26, 35.17]	1.88	[-0.09, 3.85]	1.060	[0.999, 1.125]	.175
Product modification	361	32.41	[27.79, 37.40]	1.10	[-3.73, 5.94]	1.035	[0.892, 1.202]	.651
No reported action (reference)	117,523	31.31	—	—	—	—	—	—

Note. Action types are not mutually exclusive, so a record reporting more than one type appears in more than one comparison. The reference group always contains only records with no valid action code. Holm's step-down

correction is applied across the five comparisons. The smallest expected cell count is 113, so the chi-square approximation is sound and Fisher's exact test was not required.

Decision. Compare action types individually under RQ3; material modification and operating practice or training are the only types that separate from the no-action group, so they are the priorities for closer scrutiny and candidate predictors for the RQ4 screening model. All observed effects remain small and unadjusted, and the adjusted RQ2 model keeps any reported action as its exposure.

Hazard Is Negligible; Clustering Demands Robust Errors

[Figure A7](#) compares rates by hazard classification. Carcinogens reduce at 31.53% [31.06, 32.01] against 31.46% [31.15, 31.77], a negligible observed difference of 0.07 percentage points. PBT chemicals reduce slightly more often, 32.46% [31.80, 33.14] against 31.31% [31.03, 31.59], about 1.1 points, intervals only just separating. The clear exception is dioxins: 24.84% [22.73, 27.07] against 31.56% [31.30, 31.82], roughly 6.7 points lower on 1,518 records. Reported in grams rather than pounds, they are analyzed separately as a sensitivity check. [Figure A12](#) confirms prevalence and release distributions were profiled on action-year information only.

Finally, these are not independent observations. The 122,017 modeling records come from only 14,836 facilities, averaging 8.22 records each, median 4 and maximum 210; 92.8% come from facilities contributing more than one ([Figure A3](#)). Treating them as independent would understate standard errors (Cameron & Miller, 2015), so facility-clustered errors are a requirement of the RQ2 specification, not a refinement. [Figure A13](#) summarizes 15 data-quality checks: eight passed without corrective action, five identified issues that were controlled analytically, and two, cross-year attrition of 7.6% to 9.5% ([Figure A8](#)) and TRI's coverage limits, remain limitations of the source data.

Decision. Analyze dioxins separately, and cluster standard errors by facility as a requirement of the RQ2 specification, not a refinement.

Key Business Insights from the EDA

The following insights rank what the exploratory data show for an environmental health and safety (EHS) audience. All rest on observed, unadjusted rates, none establishes causality, and each is revisited under the relevant research question once the adjusted models are fitted.

1. Reported prevention actions show a small positive difference. The 3.61-point raw gap ([Figure 4](#)). The direction is encouraging, but the magnitude is modest and the two groups differ on seven of 19 covariates, so this is a signal to test rather than a result to bank.

2. Material modification and operating practice or training show the strongest preliminary results. Only these two separate from the no-action group after Holm correction ([Figure 6](#), [Table 1](#)). If any actions warrant priority attention, these are the candidates.

3. Industry type has a meaningful influence on reduction rates. Rates span 15.4 points across the 15 qualifying subsectors ([Figure 3](#)), and the subsectors reporting the most actions are not those reducing most often. Where a facility sits matters at least as much as what it reports.

4. Production change and current release level must be considered before interpreting action effectiveness. Both confound the raw comparison ([Figure 5](#); [Figure A6](#)), patterns consistent with production swings and regression to the mean rather than prevention. Any action comparison that ignores these will overstate what prevention achieves.

These rankings frame the modelling that follows; they are working hypotheses, not conclusions.

Modelling

This section states the design, the technique chosen for each research question and why, the features entering the models, and the metrics results will be judged by. No final multivariable RQ2 model or predictive RQ4 model has been fitted yet; those analyses begin in Week 5. Preliminary univariable models were fitted only to assess functional form during EDA. Specifying metrics in advance removes the temptation to pick the measure that flatters the result.

Overall Design

The design is a retrospective observational study with a temporal, out-of-time split: action years 2020 to 2022 form the development set and 2023 the reserved holdout. This mirrors real use—an EHS team has this year’s filings and wants to know about next year—so the model is trained on the past and tested on an unseen future. A random split would leak across time, because facilities recur and would land on both sides.

Holdout protection is implemented as code, not intention: all construction of the next-year outcome happens inside one function, so intermediate objects touching 2024 releases go out of scope on return. The reserved dataset is then checked by assertion for zero outcome-bearing columns and all 23 required non-outcome columns; that check passes, and release summaries elsewhere exclude 2024.

[Figure A14](#) summarizes the complete workflow from data acquisition to final holdout evaluation, separating data preparation, explanatory analysis, predictive model development, and final testing. All transformations are estimated on development-training data only, and the reserved 2023-to-2024 dataset is used once, after the feature set, model specification, hyperparameters, and decision threshold are locked.

Choice of Techniques with Justification

RQ1: Chi-Square Tests of Independence

RQ1 pairs one categorical predictor with a binary outcome, the textbook case for a Pearson chi-square test of independence; two separate tests are used because each factor is of interest on its own. Cramér's V accompanies each, since with more than 120,000 records chi-square reaches significance for effects too small to act on, and Wilson score intervals are reported for every proportion.

RQ2: Logistic Regression with Facility-Clustered Standard Errors

The binary outcome makes logistic regression the natural choice, with coefficients that convert into interpretable odds ratios. Two data features force refinements: repeated records per facility require facility-clustered standard errors (Cameron & Miller, 2015), and the curved production ratio is carried as a centered quadratic term. Random forests were rejected because RQ2 asks about one coefficient and its confidence interval, which a tree ensemble does not provide.

RQ3: Two-Proportion Comparisons with Holm Correction

Each action type is compared with the no-action group in a 2×2 table, using Pearson chi-square where expected counts allow and Fisher's exact test otherwise; the smallest expected count is 113, so the approximation holds. Because five comparisons share one reference group, Holm's (1979) correction controls family-wise error with more power than Bonferroni, and rate differences and relative risks with Katz log-scale intervals accompany the p-values.

Because facilities contribute repeated records, the primary proportion comparisons are supplemented by facility-clustered binary-regression sensitivity checks to confirm the RQ1 and RQ3 conclusions hold after accounting for within-facility dependence.

RQ4: Regularized Logistic Regression Against a Simple Baseline

An elastic-net penalized logistic regression is planned because it preserves interpretability, guards against overfitting given 15 subsector and five correlated action indicators, and degrades gracefully when predictors are weak. It is compared against a simple baseline using only the action indicator and current release, so added complexity must earn its place.

The RQ4 model omits categorical action-year indicators because the reserved test contains action year 2023, a category unseen in training. A secondary development-only specification may add a numeric time index, retained only if it improves cross-validated performance consistently across transitions without introducing an unseen category in the holdout.

Model tuning will use leave-one-transition-out validation within the development period, training on two transitions and validating on the third. The elastic-net model will be compared with the simple baseline using differences in ROC AUC and Brier score, with paired bootstrap confidence intervals resampled by facility to judge whether the difference is practically distinguishable from zero. The final model will be refitted on all development transitions before the single reserved-test evaluation.

Features and Feature Engineering

Categorical subsector and action-year variables use one-hot indicators with prespecified reference categories; Chemical Manufacturing (NAICS 325) is the RQ2 reference subsector, and

below-threshold subsectors form an explicit Other Manufacturing category rather than being absorbed into the reference. Every continuous transformation—standardization, imputation, capping, centering, and dummy construction—is fitted inside the training fold and applied unchanged to validation data.

Feature roles were fixed in code before any data were read, so identifiers cannot leak in as predictors. Four identifier columns—facility identifier, chemical identifier, Chemical Abstracts Service (CAS) number, and chemical name—are carried for traceability only and never offered to a model. Fifteen predictor and four audit columns complete the schema, with an assertion that no column appears in both lists. [Table E7](#) lists every feature and where it will be used.

Evaluation Metrics

[Table E8](#) lists every metric that will be reported, with its formula. Discrimination, calibration, and error are all represented, following Steyerberg et al. (2010) and Van Calster et al. (2019), because a model that ranks well but is poorly calibrated will still mislead anyone who acts on its probabilities. Because no model has yet been fitted, the table reports formulae only; the corresponding worked calculations will accompany the final results.

Preliminary Results

RQ1: Differences Across Years and Subsectors

RQ1 is substantially answered: reduction rates differ across the three transitions and the 15 subsectors, rejecting both null hypotheses. But both effects are weak—year and industry shift the expected probability, not enough to screen on alone.

RQ2: Adjusted Association

RQ2 is deliberately not yet answered: the unadjusted 3.61-point advantage cannot stand when seven of 19 covariates are imbalanced. The design matrix is full rank at 24 of 24 columns, multicollinearity is limited (maximum variance inflation factor 2.06, condition number 80.7), and the empirical logit generally increases with log current release. Remaining nonlinearity will be examined when the full model is fitted.

RQ3: Action Types Individually

RQ3 is answered at the unadjusted level in [Table 1](#): two of five action types differ from the no-action group after Holm correction, so those nulls are rejected and three retained. Effect sizes are very small and unadjusted; that material modification leads is consistent with Cagno et al. (2005).

RQ4: Predictive Screening

RQ4 has not been tested and will not be until the model is locked, but a fair test is possible: the reserved dataset holds 39,966 eligible feature-only records, verified to carry all 23 required non-outcome columns and no outcome column. The primary evaluation will use its pound-reported subset, matching the development population, with exact counts and candidate-parameter numbers reported once the final modelling frame is generated (Peduzzi et al., 1996; Riley et al., 2019; van Smeden et al., 2019). Given the weak effects throughout, only a modest area under the curve is expected.

Interim Limitations and Risks

Each limitation is paired with its planned mitigation.

- The outcome cannot separate prevention from production change. Mitigation: the production ratio is a control, and findings are reported as associations, never causal effects.
- Some facility-chemical records are absent the following year (cross-year attrition of 7.6% to 9.5%), and TRI covers neither all chemicals nor all facilities. Disappearance may reflect a reduction below the reporting threshold, closure, substitution, operational change, or reporting differences, and the direction of the resulting selection bias cannot be established. Mitigation: matched and unmatched records are profiled by transition and action status, unmatched records are not treated as zero releases or reductions, and claims are limited to TRI-reporting manufacturing.
- A zero action indicator means no valid source-reduction code was observed in the selected File 2A fields; it does not prove the facility took no action. Unreported or inconsistently reported actions may weaken or distort the observed association in a direction that cannot be determined. Mitigation: results are described as associations with reported actions, not effects of all prevention activity.
- The 2020–2021 transition carries both a pandemic disruption and the coding-scheme change. Mitigation: the sensitivity analysis excluding it is mandatory.
- Regression to the mean mimics prevention ([Figure A6](#)). Mitigation: current release level is a required control in every model.

Next Steps

- Week 5. Fit the RQ2 adjusted logistic regression with facility-clustered standard errors and report adjusted odds ratios.

- Week 6. Run both sensitivity analyses and test whether the provisional quadratic ratio term survives the full model.
- Week 7. Build, cross-validate and lock the RQ4 elastic-net screening model and its simple baseline.
- Week 8. Evaluate once on the reserved holdout, report discrimination and calibration, and set a screening threshold.

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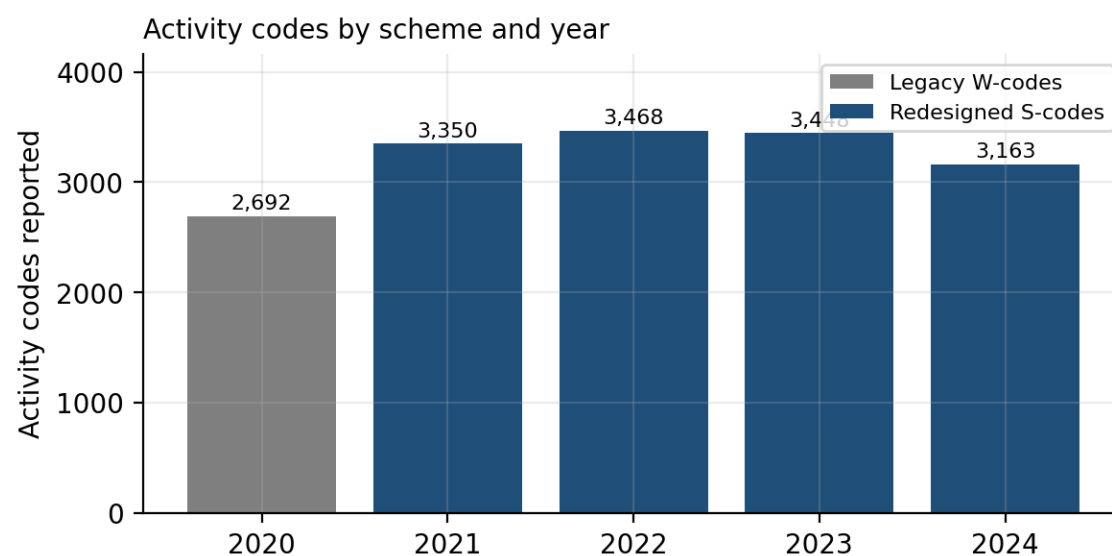
Appendix A

Exploratory Data Analysis Figures

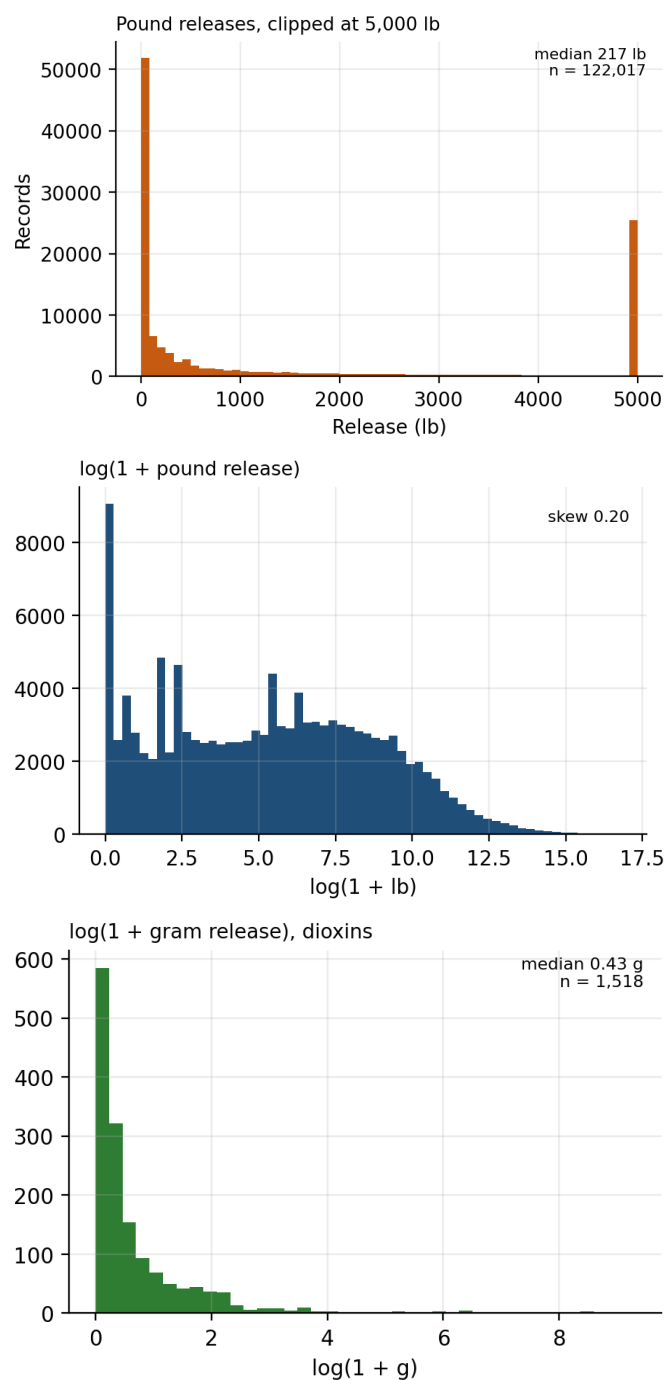
Every figure below was produced by the notebook in the project repository. Each is referenced in the Analysis section of the main text.

Figure A1

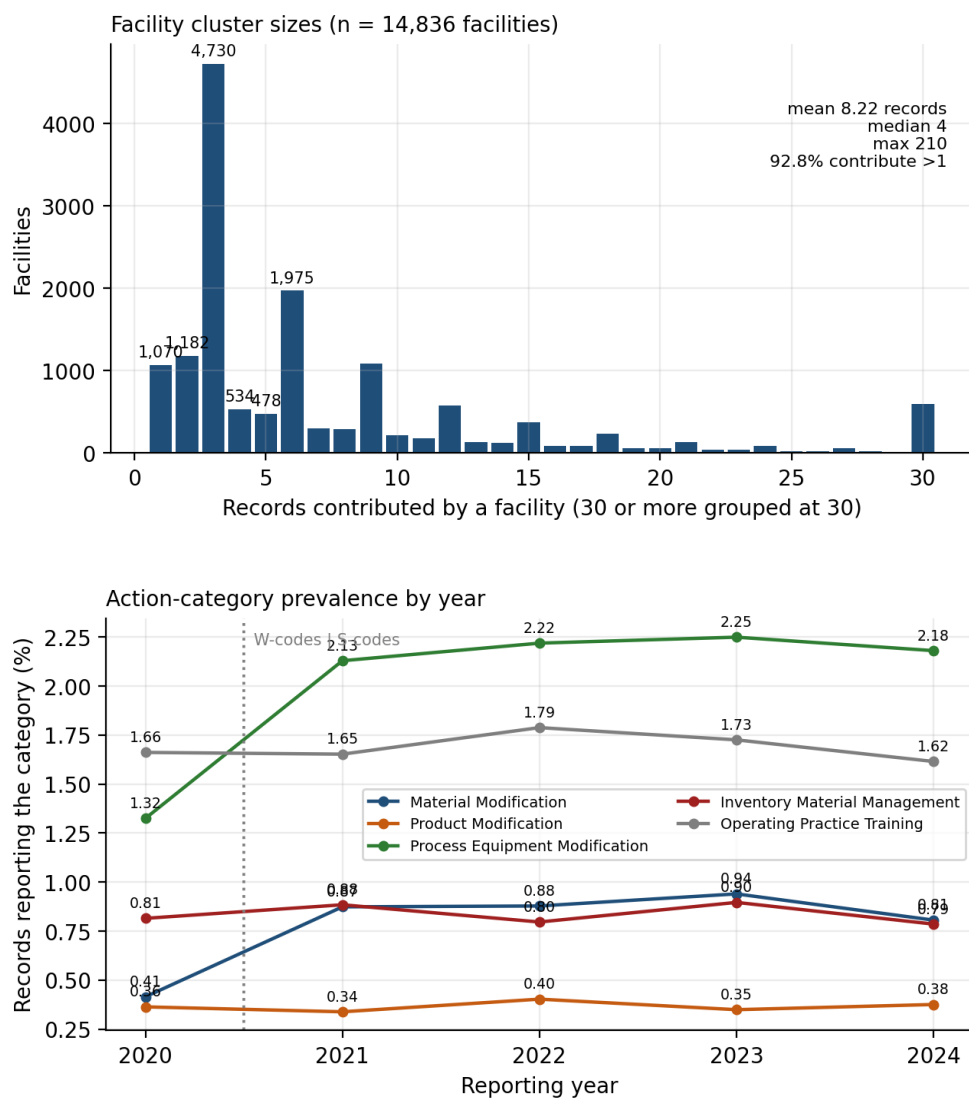
Source-Reduction Activity Codes by Coding Scheme and Reporting Year



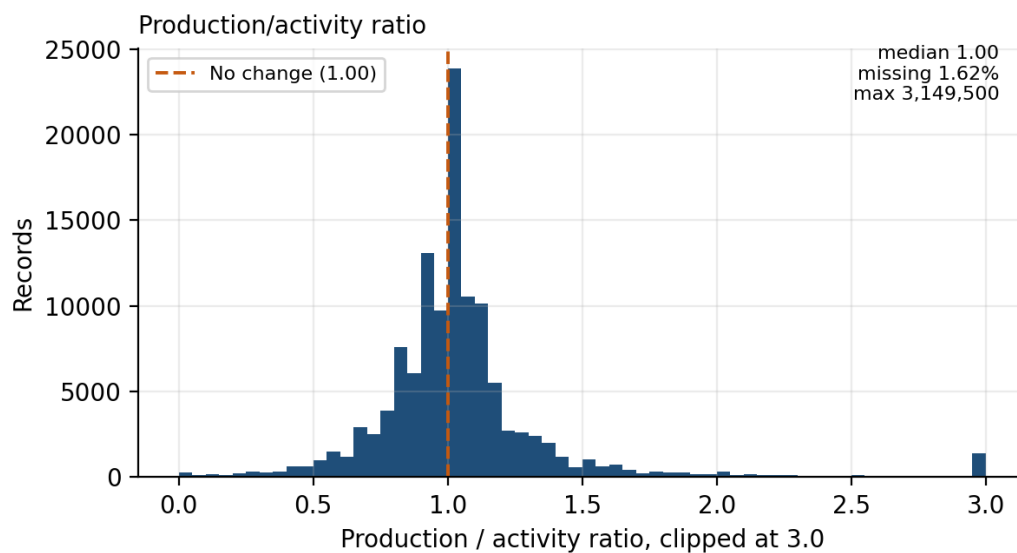
Note. Legacy W codes appear almost exclusively in 2020 and are replaced by S codes from 2021 onward. All 70 distinct codes map to the five prespecified action types, with zero unmapped codes asserted in the pipeline.

Figure A2*Distribution of Current-Year Releases*

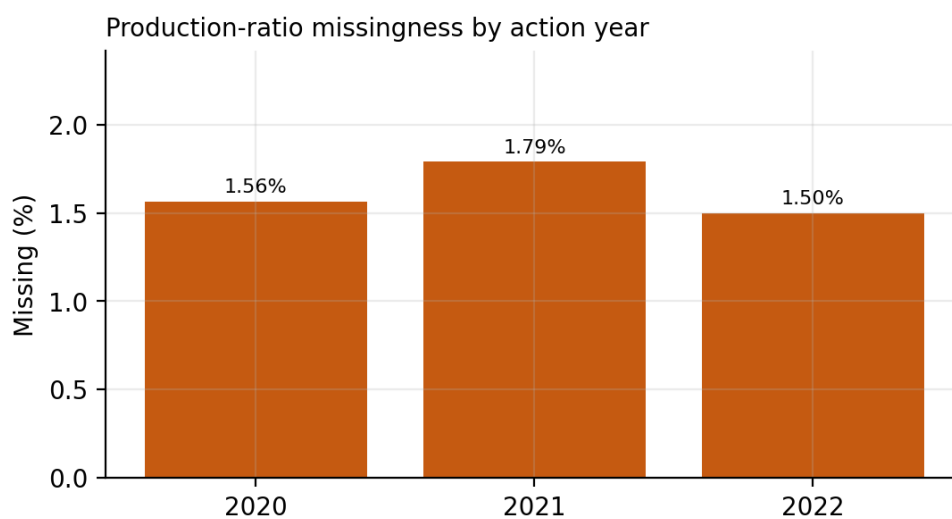
Note. Pound-reported releases are shown clipped at 5,000 lb and on a log scale; gram-reported dioxins are shown separately. Skewness is 34.6, with a median of 217 lb against a mean of 24,737 lb.

Figure A3*Facility Cluster Sizes and Action-Category Prevalence by Reporting Year*

Note. The top panel shows the distribution of records contributed per facility (mean 8.22, median 4, maximum 210), with 92.8% of records coming from facilities that contribute more than one. The bottom panel shows action-category prevalence by reporting year: process or equipment modification rises from 1.32% in 2020 to 2.13% in 2021, coinciding with the coding-scheme change, while product modification, inventory management, and operating practice remain flat across the same boundary.

Figure A4*Distribution of the Production or Activity Ratio*

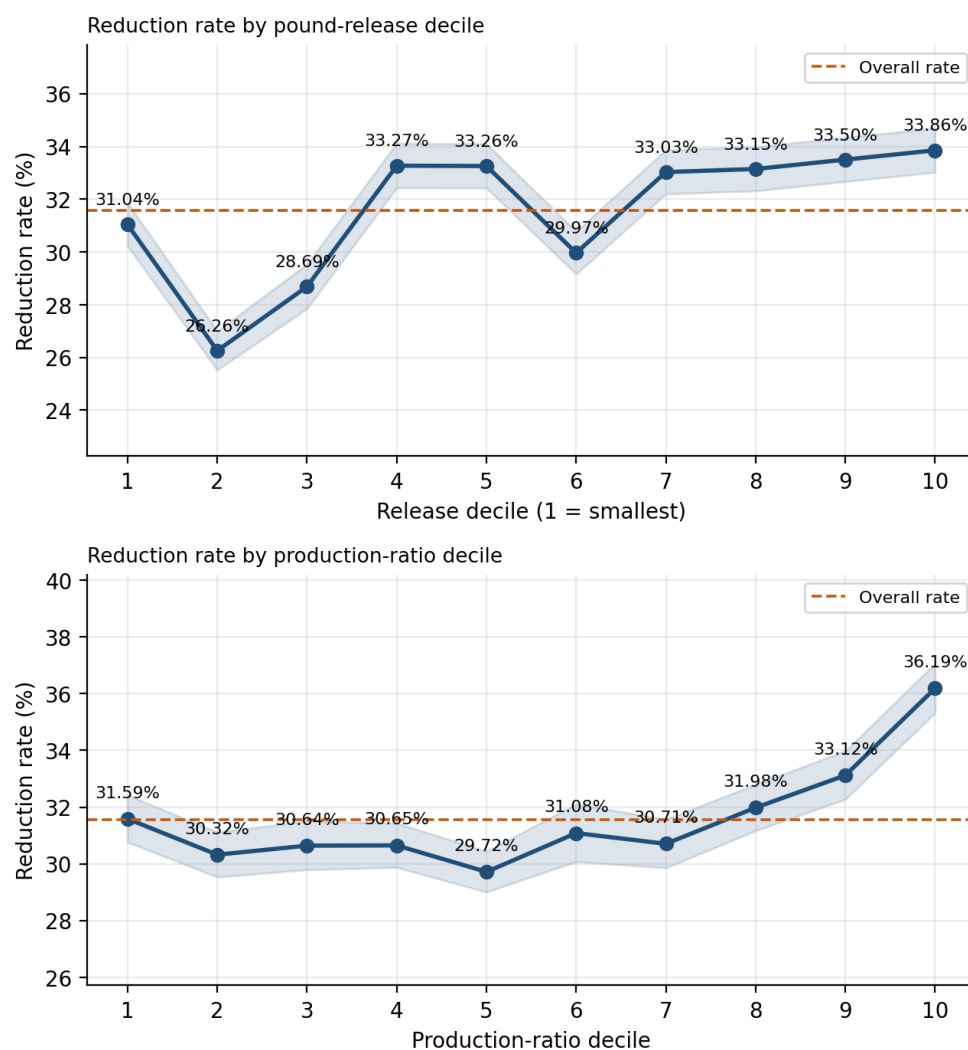
Note. The ratio is centered on 1.00, meaning most facilities report production similar to the prior year. Extreme values above three million are treated by training-fold capping before centering and squaring rather than entering the model unchanged.

Figure A5*Production-Ratio Missingness by Covariate Group*

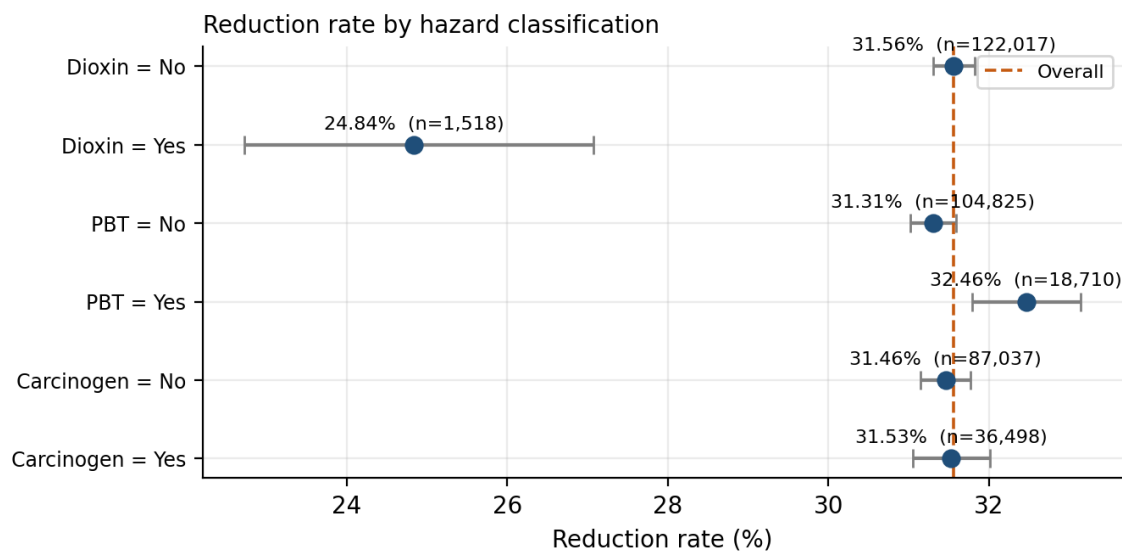
Note. Production-ratio missingness remains low across the three development action years, ranging from 1.50% to 1.79%. Additional comparisons by reported-action status, hazard classification, and outcome are reported in the EDA notebook; [Table E6](#) summarizes the resulting modelling decision. The missingness mechanism cannot be confirmed from observed data.

Figure A6

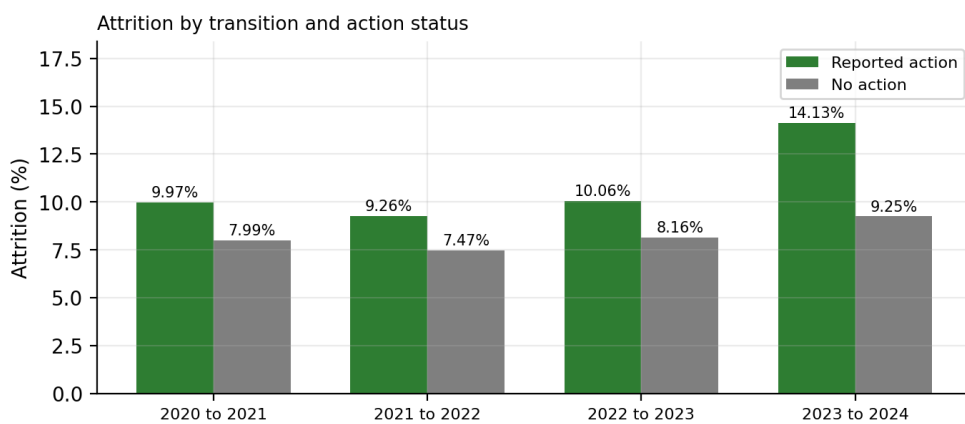
Reduction Rate by Release Decile and Production-Ratio Decile

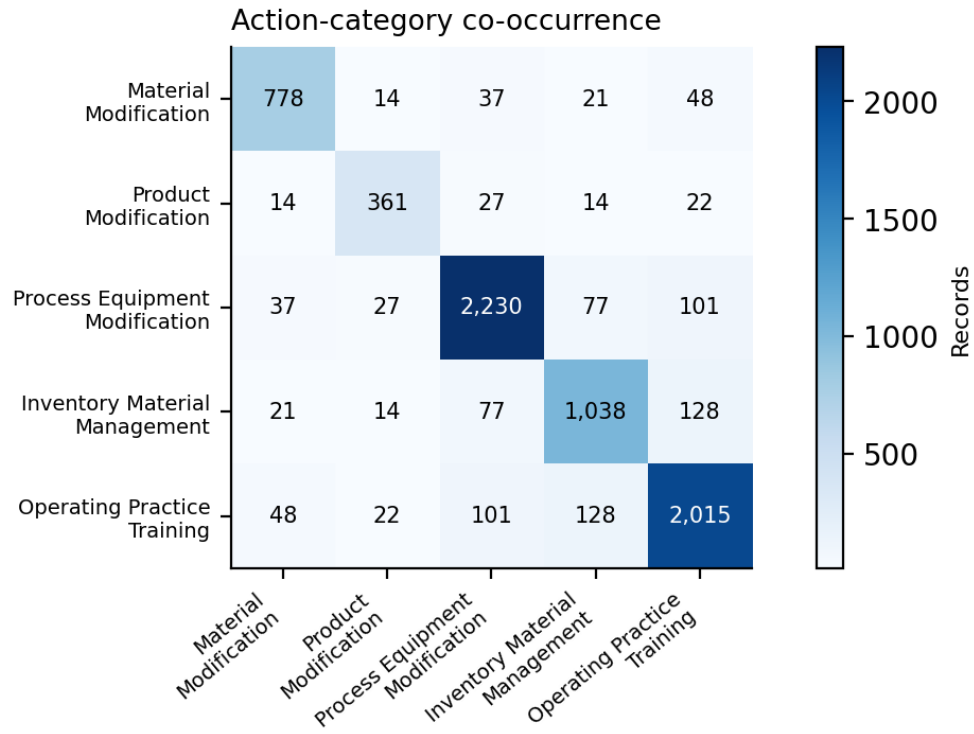


Note. Reduction rates vary across current-release deciles and are generally higher in several upper deciles, although the pattern is not monotonic. The production-ratio relationship is also non-monotonic.

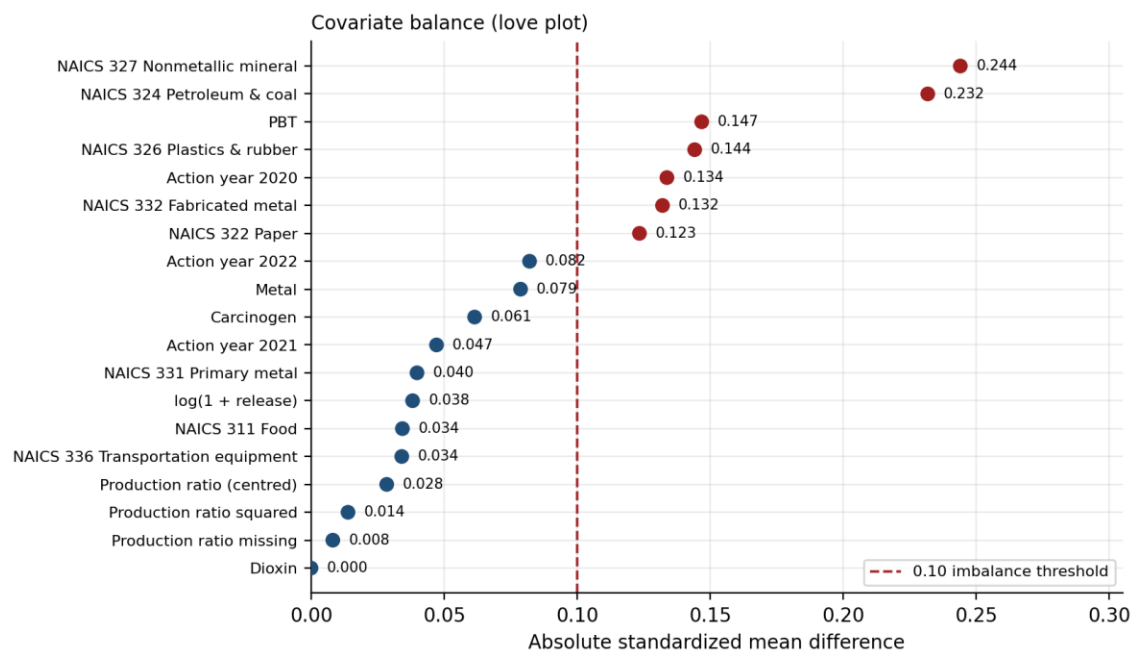
Figure A7*Reduction Rate by Chemical Hazard Classification*

Note. Carcinogen status shows a difference of 0.07 percentage points. Dioxins reduce 6.7 points less often than other chemicals and are analyzed separately.

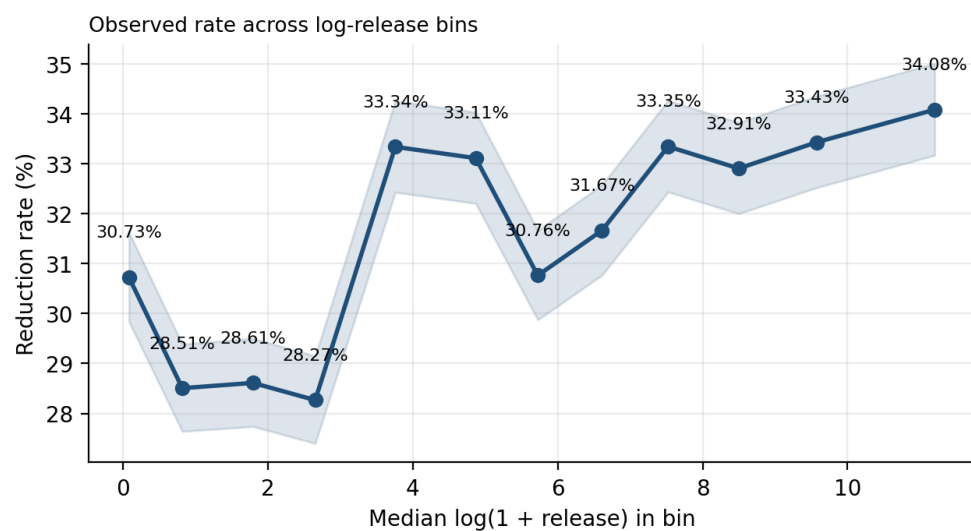
Figure A8*Cross-Year Attrition and Action-Category Co-Occurrence*

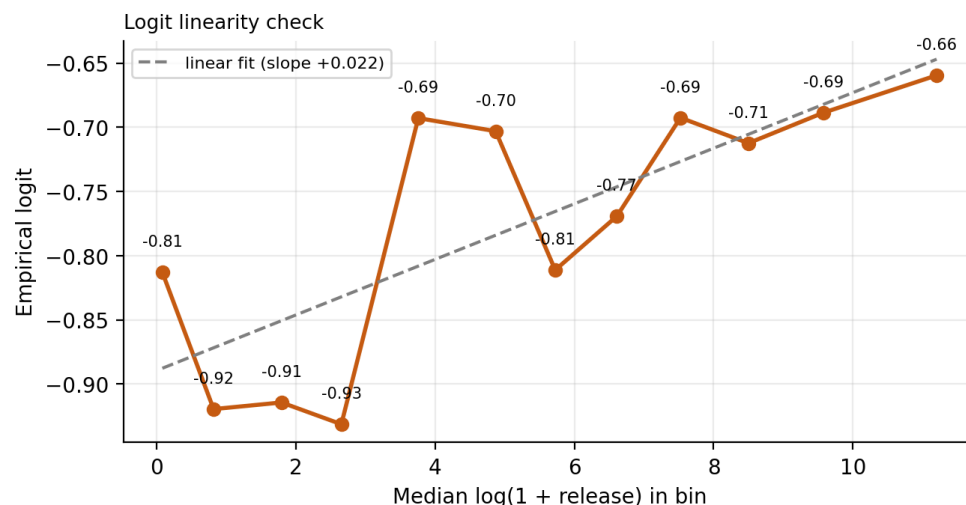


Note. Overall attrition ranges from approximately 7.6% to 9.5% across transitions. Within the action-status groups shown here, attrition ranges from 7.47% to 14.13%, with the highest value among action-reporting records in the 2023-to-2024 transition. Records reporting an action average 1.07 distinct action categories, so co-reporting is uncommon.

Figure A9*Covariate Balance Between Action and No-Action Groups*

Note. Seven of 19 covariates exceed a standardized mean difference of 0.10. The largest absolute standardized mean differences are 0.244 and 0.232, both reflecting differences in industry composition.

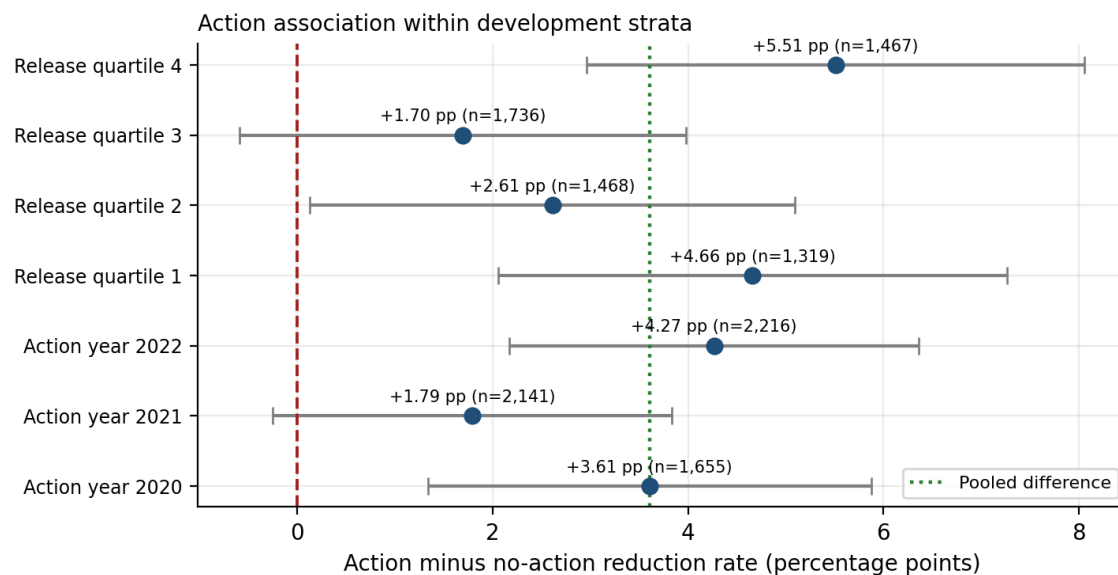
Figure A10*Observed Reduction Rate Across Log-Release Bins with Logit Linearity Check*



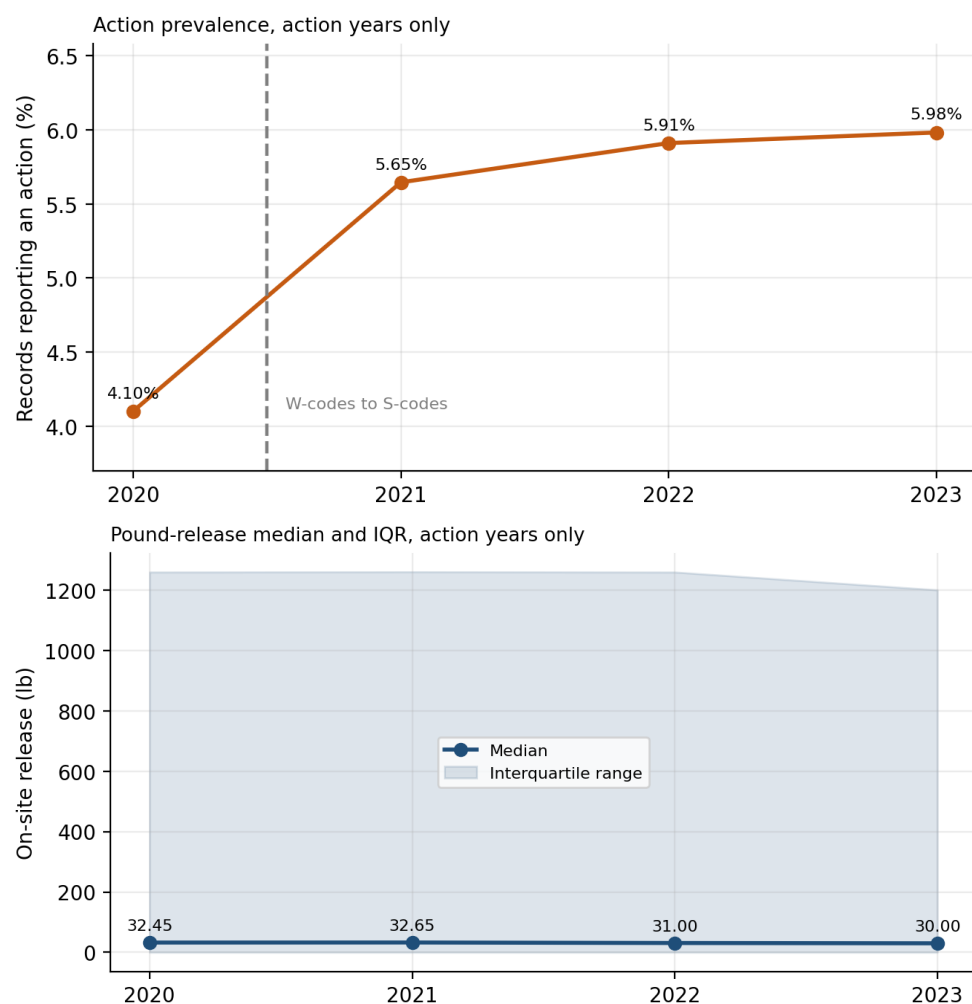
Note. The empirical logit generally increases with log current release, supporting the use of a log transformation. Some departures from linearity remain, so the linear log-release term will be checked against a flexible alternative during development-model diagnostics.

Figure A11

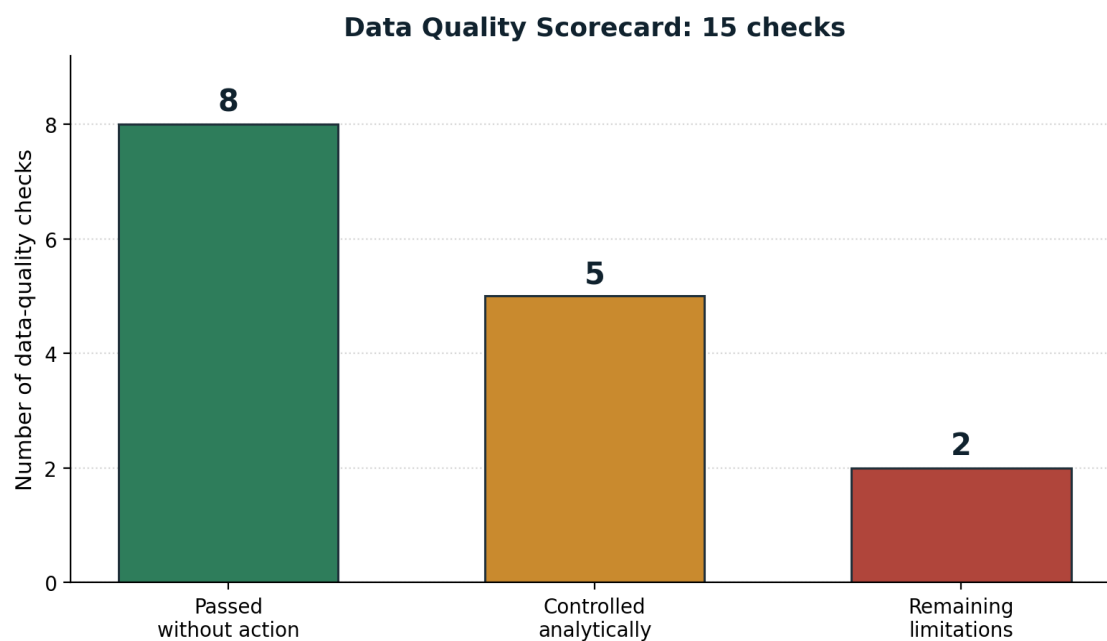
Action Association Within Development Strata



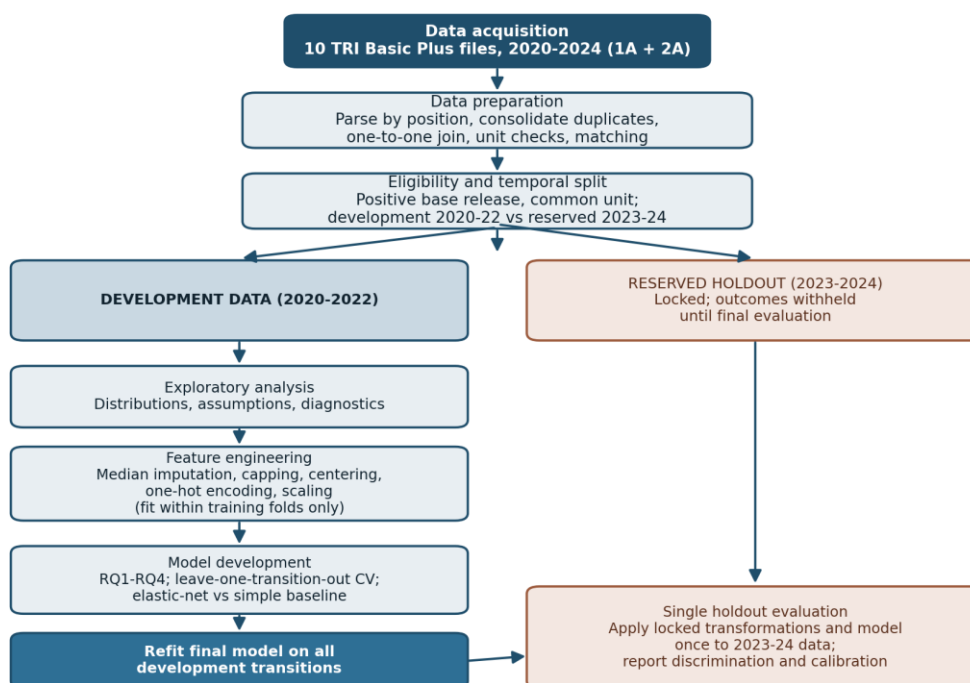
Note. The point estimates are positive across the displayed development strata, including after excluding the 2020-to-2021 transition. Several confidence intervals include zero, so these are exploratory consistency checks rather than separate evidence of significance.

Figure A12*Action Prevalence and Release Distribution, Action Years Only*

Note. Reported action prevalence and the median and interquartile range of pound releases, restricted to action years so that no outcome-year information is used. [Figure A2](#) describes eligible positive-release development records, while [Figure A12](#) uses consolidated action-year records before the positive-base-release eligibility restriction. The medians therefore summarize different analytical populations and should not be compared directly.

Figure A13*Data Quality Scorecard Outcomes*

Note. [Figure A13](#) summarizes 15 data-quality checks: eight passed without corrective action, five identified issues that were controlled analytically, and two remain limitations of the source data.

Figure A14*End-to-End Data Preparation, Analysis, and Evaluation Workflow***Appendix B****Source File Provenance and Repository Structure**

All ten checksums are distinct, confirming that no file was accidentally duplicated during download. Any reader can verify their own copies against the values below.

Table B1*Source File Sizes and SHA-256 Checksums*

Reporting year	File type	Size (MB)	SHA-256 (first 16 characters)
2020	1A	62.6	b94f8cb1286d7420
2020	2A	44.5	5ddaa1624fbba793
2021	1A	62.2	73daa53931a58696

Reporting year	File type	Size (MB)	SHA-256 (first 16 characters)
2021	2A	44.4	805fbdc932f51aef
2022	1A	63.6	4cf809c08d11cd8c
2022	2A	45.4	16f382966622b5e8
2023	1A	62.6	0c60fa25c4ac4348
2023	2A	44.7	cc3017cd0ca0b310
2024	1A	61.6	f463ec07ff320968
2024	2A	44.0	dbedc8bafc05338a

Note. Checksums are truncated for readability. Full 64-character values are printed by the notebook and stored in the repository.

Table B2

Consolidation Counts by Reporting Year

Year	Before	After	Removed	Groups with duplicates	Revision superseded	Summed originals
2020	57,004	56,462	542	321	33	288
2021	56,750	56,219	531	353	39	314
2022	56,725	56,155	570	367	13	354
2023	55,882	55,349	533	333	26	307
2024	54,595	54,115	480	304	16	288

Note. Unit conflicts and chemical-identifier conflicts were zero in every year. Summing dominates supersession, which is why the original tie-break rule was amended (see Appendix C).

Repository Structure

The public repository is organized as follows:

- README.md — problem statement, data source, and reproduction instructions.
- data/ — data dictionary, download instructions, checksums, and a 200-row sample for each reporting year.
- notebooks/ — the exploratory data analysis notebook that produced every figure in Appendix A.

- scripts/ — the data-preparation pipeline, including parsing, consolidation, matching, and export.
- docs/ — the synopsis and this interim report.

Appendix C

Proposed Amendment to the Synopsis Duplicate-Handling Rule

The synopsis specified that when a facility files more than one Form R for the same chemical in the same reporting year, the authoritative release quantity should be taken from the filing with the highest document control number. Implementation showed this rule to be incorrect for the dominant case, and this appendix records the amendment and the evidence for it.

Duplicate filings arise for two different reasons. In the first, a facility submits a revised form that replaces an earlier one; here the revision genuinely supersedes the original, and taking one filing is correct. In the second, and much more common, a facility submits several original forms that each carry part of the year's total for that chemical; here taking a single filing discards real quantity and understates the release. Across the five reporting years, 1,551 duplicate groups were of the second kind and only 127 were of the first.

The amended rule is therefore conditional. If any filing in a facility-chemical-year group carries a value in either of the two revision-code fields, the revised filing supersedes and the originals are discarded. Otherwise, the release quantities of all filings in the group are summed. The sum uses a minimum-count setting of one, so that a group in which every value is missing produces a missing total rather than a false zero. Unit consistency is checked within each group before any summation, and the release is set to missing if more than one unit appears; no such conflict occurred in any year.

When one or more revision-coded filings exist within a facility-chemical-year group, the revision-coded filing with the highest document control number supplies the authoritative release

quantity and unit. Earlier revisions and original filings are treated as superseded. Source-reduction indicators are taken from the authoritative filing; the production or activity ratio is taken from that filing when available. When no revision-coded filing exists, the release quantities of the original filings are summed after unit and chemical-identifier consistency checks. This makes the rule deterministic even when several original or revised filings occur in the same group.

Two honest notes accompany this amendment. First, adding the second revision-code field and the minimum-count safeguard changed no results: revision counts were identical with one field or two, and no group consisted entirely of missing values. Both were retained as defensive measures rather than as corrections. Second, the synopsis remark that duplicate original forms represent separate process units has been withdrawn, because the TRI fields available in this study cannot establish that. The forms are summed because summing preserves the reported total, which is a defensible rule regardless of why the duplicates exist.

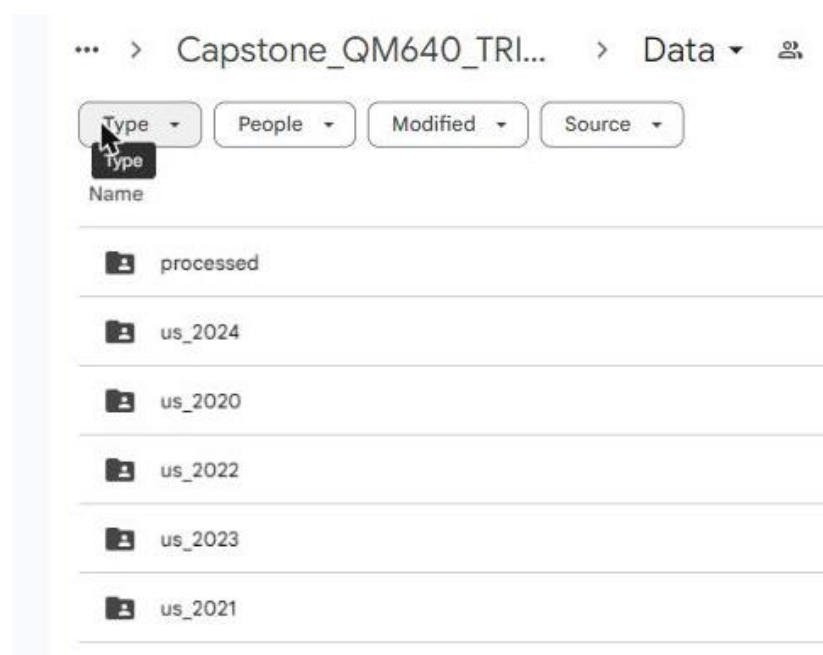
Appendix D

Data Availability Evidence

The screenshots below evidence the folder structure and dataset contents described under Data Availability. Access is set to anyone with the link, viewer permission.

Figure D1

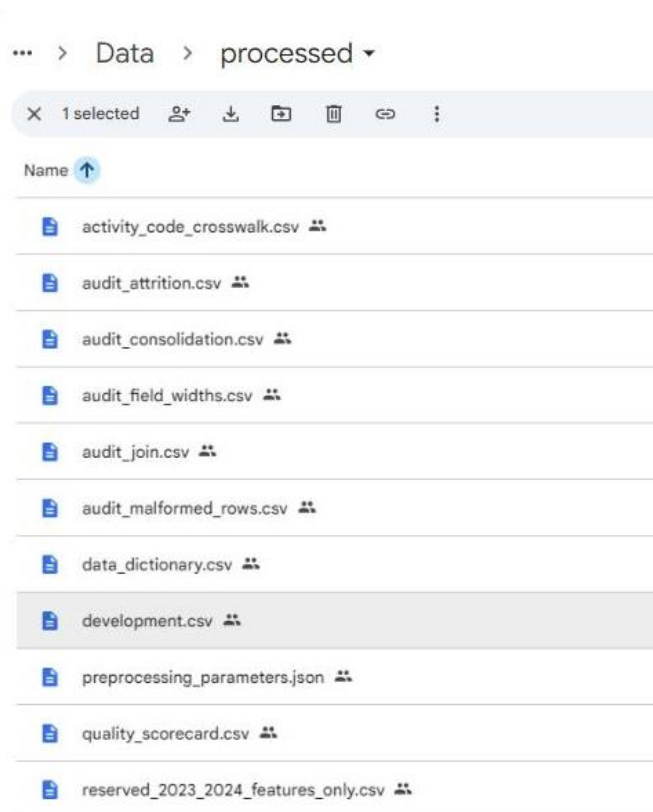
Raw Data Folder Structure on Google Drive



Note. One folder per TRI reporting year, 2020 through 2024, each holding File 1A and File 2A, plus the processed subfolder. The folder icons carry the shared badge.

Figure D2

Processed Data Folder Contents



Note. Eleven files. The holdout (4.3 MB) is smaller than the development dataset (16.1 MB) because it carries no outcome columns.

Figure D3

Preview of the Processed Development Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	activity_id_new	chemical_id_new	cas_number_new	chemical_name_subsector_new	year	release_new	production_ratio_per_ratio_mining	is_carcinogen_new	is_pbt_new	is_dfoah_new	is_hazard_new	reported_why_action	material_modification	product_modification	process_modification	process_modification	process_modification
2	15301PLUPP0000	N130		Dioxypentenes	326 2020_2021	25	1	0	0	0	0	0	0	0	0	0	0
3	726017NMT210N	100426	100-42-5	Styrene	326 2020_2021	39540	0.9	0	1	0	0	0	0	0	0	0	0
4	99220PNSBUJLD	7440473	7440-47-3	Chromium	331 2020_2021	77.36	1	0	0	0	0	1	0	0	0	0	0
5	99220PNSBUJLD	7439965	7439-96-5	Manganese	331 2020_2021	248.71	1	0	0	0	0	0	0	0	0	0	0
6	99220PNSBUJLD	7440020	7440-02-0	Nickel	331 2020_2021	105.08	1	0	1	0	0	1	0	0	0	0	0
7	37398CKNDRT3E	100414	100-41-4	Ethylbenzene	326 2020_2021	699	0.22	0	1	0	0	0	1	1	1	0	0
8	37398CKNDRT3E	106883	106-88-3	Toluene	326 2020_2021	4239	0.15	0	0	0	0	0	1	1	1	0	0
9	37398CKNDRT3E	1306597	1306-59-7	Xylene (mixed isom)	326 2020_2021	644	0.21	0	0	0	0	1	1	1	0	0	0
10	76146MOLN4600E	7440508	7440-50-8	Copper	333 2020_2021	118	1.06	0	0	0	0	1	1	1	0	0	1
11	76146MOLN4600E	N430		Lead compounds	331 2020_2021	33	1.04	0	0	1	0	1	1	1	0	0	0
12	28644ILYK101CA	N230		Certain glycol ethers	325 2020_2021	5	0.84	0	0	0	0	0	0	0	0	0	0
13	28644ILYK101CA	50000	50-00-0	Formaldehyde	325 2020_2021	156.99	1.02	0	1	0	0	0	0	0	0	0	0
14	28644ILYK101CA	7697372	7697-37-2	Nitric acid	325 2020_2021	5	1.25	0	0	0	0	0	0	0	0	0	0
15	14PLKCR15NF	106852	106-85-2	Phenol	332 2020_2021	11148	0.79	0	0	0	0	0	0	0	0	0	0
16	64WRSC4L2CUJ	7440473	7440-47-3	Chromium	333 2020_2021	0.45	0.76	0	0	0	0	1	0	0	0	0	0
17	4630RSC4L2CUJ	7440508	7440-50-8	Copper	333 2020_2021	0.59	0.76	0	0	0	0	1	0	0	0	0	0
18	4630RSC4L2CUJ	7440020	7440-02-0	Nickel	333 2020_2021	0.35	0.76	0	0	0	0	1	0	0	0	0	0
19	6430BLVNS9WY71	181242	181-24-2	Benzo[a,h]pyrene	324 2020_2021	0.01		1	0	1	0	0	0	0	0	0	0
20	6561VBLVNS9WY71	N590		Polycyclic aromatic	324 2020_2021	2000		1	1	1	0	0	0	0	0	0	0
21	68901DTNL601W	7439921	7439-92-1	Lead	332 2020_2021	0.3	0.98	0	1	1	0	1	0	0	0	0	0

Note. Column order follows the schema declared in code: four identifier fields, then predictors, then audit fields.

The span column records the action-year to outcome-year transition. Values shown are real records from reporting year 2020.

Figure D4

Activity Code Crosswalk with EPA Descriptions

	A	B	C	D	E	F	G	H	I	J
1	Code	Category	EPA description	2020	2021	2022	2023	2024	Total	Status
2	901	material_modificatio	SUBSTITUTED A FL	0	29	22	26	10	87	mapped
3	902	material_modificatio	SUBSTITUTED AN C	0	36	41	31	28	136	mapped
4	903	material_modificatio	SUBSTITUTED NAA	0	177	168	185	180	690	mapped
5	904	material_modificatio	SUBSTITUTED NAA	0	83	81	92	81	337	mapped
6	905	material_modificatio	MODIFIED CONTEN	0	68	69	41	34	212	mapped
7	906	material_modificatio	OTHER MATERIAL I	0	102	117	149	132	500	mapped
8	911	product_modificatio	REFORMULATED O	0	98	111	90	90	389	mapped
9	912	product_modificatio	ALTERED DIMENSIO	0	21	31	30	27	109	mapped
10	913	product_modificatio	MODIFIED PRODUCT	0	10	7	5	8	30	mapped
11	914	product_modificatio	OTHER PRODUCT I	0	61	77	68	78	284	mapped
12	921	process_equipment	OPTIMIZED PROCE	0	565	577	545	632	2420	mapped
13	922	process_equipment	INSTITUTED RECI	0	38	57	46	43	184	mapped
14	923	process_equipment	IMPLEMENTED NEV	0	148	150	115	122	535	mapped
15	924	process_equipment	MODIFIED OR UPDA	0	204	190	233	195	822	mapped
16	925	process_equipment	OTHER PROCESS I	0	259	286	214	198	957	mapped

Note. Every observed code is mapped to one of the five action types, with counts by year and a status column reading mapped throughout. Legacy W codes carry counts only in 2020; S codes carry counts only from 2021 onward, evidencing the scheme change discussed in the Analysis section.

Appendix E

Supporting Tables

These tables are referenced from the main text. They are placed here because they are reference material rather than analytical results, keeping the body of the report within the required length.

Table E1

Literature Relevance Matrix

Author (Year)	Domain / Setting	Method	Key finding	How it supports this project
Hamilton (1995)	TRI disclosure, U.S. public firms	Event study of stock returns	Firms named in the first TRI release lost significant market value on the announcement.	Establishes that TRI reporting carries real consequences, so reported actions are worth studying.
Konar & Cohen (1997)	Community right-to-know, U.S. facilities	Panel comparison of emissions	Firms with the largest stock-price losses cut emissions most in later years.	Supports the premise that reported behavior and later releases are linked.
Gamper- Rabindran (2006)	TRI program effect, U.S. manufacturing	Regression discontinuity	Reported reductions were partly reporting changes, not real abatement, and varied by industry.	Direct warning behind this study's refusal to claim causality, and behind the subsector analysis in RQ1.
Ranson et al. (2015)	Pollution prevention, U.S. manufacturing	Facility-level panel regression	Source-reduction activity was associated with meaningful declines in reported releases.	Closest prior work; motivates the RQ2 and RQ3 comparisons on recent data.
King & Lenox (2002)	Waste prevention vs. treatment	Panel analysis of U.S. firms	Prevention, not end-of- pipe treatment, drove the profitable share of pollution reduction.	Justifies focusing on source-reduction codes rather than treatment or recycling fields.

Author (Year)	Domain / Setting	Method	Key finding	How it supports this project
Cagno et al. (2005)	Cleaner production projects	Analysis of 134 project reports	Prevention projects differed widely in payback; equipment and process changes performed best.	Supports treating the five action types separately in RQ3 rather than pooling them.
Peduzzi et al. (1996)	Logistic regression methodology	Simulation study	Fewer than about ten events per variable produced biased coefficients.	Sets the events-per-variable floor used in the sample-size plan.
Riley et al. (2019)	Prediction model sample size	Analytical derivation	Proposed a minimum sample size based on shrinkage and precision rather than a fixed rule.	Used to check that the development set can support the planned RQ4 model.
Van Smeden et al. (2019)	Binary prediction models	Simulation beyond events-per-variable	Required sample size depends on predictor strength and outcome rate, not events alone.	Reinforces the RQ4 sample-size check and the decision to regularize.
Steyerberg et al. (2010)	Prediction model evaluation	Methodological framework	Discrimination alone is insufficient; calibration and clinical usefulness matter.	Determines the RQ4 metric set: AUC plus Brier score plus calibration.
Van Calster et al. (2019)	Model calibration	Methodological review	Poorly calibrated models mislead decisions even at good AUC.	Requires a calibration curve before any screening threshold is recommended.
Cameron & Miller (2015)	Clustered inference	Practitioner guide	Ignoring within-cluster correlation understates standard errors.	Directly justifies facility-clustered standard errors in RQ2.
Holm (1979)	Multiple comparisons	Statistical derivation	A step-down procedure controls family-wise error with more power than Bonferroni.	Applied to the five action-type comparisons in RQ3.
Agresti & Coull (1998)	Binomial interval estimation	Comparison of interval methods	The Wilson score interval outperforms the normal approximation, especially for small or extreme proportions.	All reduction-rate confidence intervals in this report use the Wilson method.
Burnham & Anderson (2002)	Model selection	Information-theoretic monograph	AIC differences below about two indicate comparable support.	Used to judge the production-ratio functional form, where the difference is 3.1.

Author (Year)	Domain / Setting	Method	Key finding	How it supports this project
EPA (2025c, 2025d)	TRI Basic Plus files and guides	Program documentation	Defines every field, code, and file layout used here.	The authoritative source for parsing positions, activity codes, and unit rules.

Note. Full bibliographic details for every source appear in the References section.

Table E2

Data Dictionary for the Analysis Variables

Variable	Definition	Type	Range or values	Missing-value handling	Role
facility_id	TRI facility identifier (TRIFD)	Text	14,851 distinct facilities	Never missing; record dropped if absent	Identifier
chemical_id	TRI chemical identifier	Text	507 distinct chemicals	Never missing	Identifier
cas_number	Chemical Abstracts Service number	Text	As reported	Retained as reported	Identifier
chemical_name	EPA chemical name	Text	As reported	Retained as reported	Identifier
release_now	Current-year total on-site releases	Numeric	> 0; max 19,750,398 lb	Zero and missing values excluded from eligibility	Predictor
release_next	Next-year total on-site releases	Numeric	>= 0	Unmatched records excluded, not set to zero	Outcome input
reduced	1 if release_next <= 0.90 x release_now	Binary	0 or 1	Defined only when release_now > 0	Outcome

Variable	Definition	Type	Range or values	Missing-value handling	Role
percent_change	Percentage change in release	Numeric	Unbounded	Defined only when release_now > 0	Outcome
production_ratio_now	Production or activity ratio, current year	Numeric	Median 1.00	Median imputation calculated within each training fold; separate missingness indicator retained; complete-case sensitivity analysis planned.	Predictor
ratio_missing	1 when the production ratio is absent	Binary	0 or 1	Derived, never missing	Predictor
subsector_now	First three digits of the primary NAICS code	Categorical	15 qualifying subsectors plus Other Manufacturing for modelling	Record dropped if outside 31-33	Predictor
span	Action-year to outcome-year transition	Categorical	Four transitions	Derived, never missing	Predictor
reported_any_action_now	1 if any valid source-reduction code is present	Binary	0 or 1	Blank codes read as 0, meaning not reported	Predictor
Five action-type indicators	Material modification, product modification, process or equipment modification, inventory or material management,	Binary	0 or 1, not mutually exclusive	Blank codes read as 0	Predictor

Variable	Definition	Type	Range or values	Missing-value handling	Role
is_carcinogen_now	operating practice or training EPA carcinogen classification	Binary	0 or 1	As supplied by EPA	Predictor
is_pbt_now	Persistent, bioaccumulative, toxic classification	Binary	0 or 1	As supplied by EPA	Predictor
is_dioxin_now	Dioxin or dioxin-like compound	Binary	0 or 1	As supplied by EPA	Predictor
is_metal_now	Metal or metal compound	Binary	0 or 1	As supplied by EPA	Predictor
unit_of_measure_now	Pounds or grams	Categorical	Pounds or grams	Record excluded if units differ across years	Audit

Note. A value of 0 on any action indicator means that no valid code was observed in the four File 2A activity-code fields. It does not prove that the facility took no environmental action.

Table E3

Retained Record Counts Against the Planned Minimum for Each Research Question

Research question	Planning basis	Minimum required	Retained	Margin
RQ1	95% confidence interval for a proportion at $p = .50$, precision .03	1,068	123,535	116x
RQ2	Power approximation, OR = 1.30, action prevalence 4.87%	10,756	122,017	11.3x
RQ3	Two-proportion comparison, 10-point difference, per action type	~290 each	361 to 2,230	1.2x to 7.7x
RQ4	20 parameters at 20 events per parameter	1,269	122,017	96x

Note. The retained sample greatly exceeds the planning minimum. The exact event-per-parameter value will be recalculated after the final pound-only feature matrix is locked.

Sample-size adequacy after cleaning. Cleaning removes slightly over half of the raw rows, so retained counts must be checked against the planning minimums rather than assumed adequate. Every requirement is met with a wide margin, and attrition against plan was negligible; [Table E3](#) gives the full comparison of retained counts against planning minimums. One synopsis concern is resolved rather than carried forward: the rarest action type was projected below the count needed, but the smallest observed group, product modification, clears it, so all five comparisons are confirmatory rather than exploratory. No research question is under-powered for the effect size it was designed to detect; whether it is powered for the effects actually observed is a different question, addressed with the RQ3 results. RQ1 and RQ3 use all eligible same-unit records because their outcome is calculated within chemical and unit. RQ2 and RQ4 use the pound-reported modelling frame because current release enters these models as a quantitative predictor. Gram-reported dioxin records will be evaluated separately in sensitivity analysis.

Table E4

Data Cleaning Log

Issue	Variables affected	Detection method	Treatment applied	Rationale
Non-Form R filings	All	Form type field	35,155 rows removed	Form A is a short certification and carries no release quantity.
Out-of-scope industries	Primary NAICS code	First two digits of NAICS	54,065 rows removed	Study scope is manufacturing (31-33); excludes mining, utilities, and waste management.
Duplicate facility-chemical forms	All	Grouping on facility, chemical, and year	2,176 rows consolidated into single records	A facility may file more than one form for the same chemical; leaving them separate would double-count.

Issue	Variables affected	Detection method	Treatment applied	Rationale
Revised filings superseding originals	Revision code fields 1 and 2	Either revision-code field non-blank	127 revised forms retained, originals discarded	A revision replaces the original submission rather than adding to it.
Genuine multiple original forms	Total on-site releases	No revision code present in the group	1,551 groups summed with min_count = 1	Summing preserves the total; min_count prevents an all-missing group from becoming a false zero.
Unit conflicts within a group	Unit of measure	More than one unit in a group	0 found; release set to missing if it occurred	Pounds and grams must never be added together.
Unit change between years	Unit of measure	Comparison across the matched pair	Record excluded from eligibility	A change of unit makes a percentage change meaningless.
Zero or missing base release	release_now	Value check	41,946 records excluded at the eligibility step	A percentage change requires a non-zero denominator.
Unmatched facility-chemical pairs	Join keys	Left join to the following year	18,738 records excluded	Without a next-year record there is no outcome; not treated as a reduction.
Missing production ratio	production_ratio_now	Numeric coercion	Replace with the training-fold median and retain ratio_missing = 1; repeat the adjusted analysis using complete cases only.	Logistic and elastic-net models require complete numeric inputs. Training-fold median imputation retains records, while the separate missingness indicator preserves information about whether the value was originally absent. Complete-case analysis will test sensitivity to this decision.
Extreme release skew	release_now	Distribution inspection	Retained; log scale used for display and diagnostics	These are real large emitters, not data errors.
Reporting-scheme change	Activity code fields	Cross-tabulation of code by year	Legacy W codes mapped to the S scheme, zero	The code scheme changed after RY2020; ignoring it would split

Issue	Variables affected	Detection method	Treatment applied	Rationale
in activity codes			unmapped asserted	one behavior across two labels.
Extreme production ratio	production_ratio_now	Percentiles and maximum-value review	Cap using training-fold limits before centering and squaring; retain original value for audit	Prevents isolated extreme values from dominating the logistic model

Table E5*Record-Count Reconciliation From Raw Files to Analysis Sets*

Stage	Records	Removed	Reason
Raw File 1A rows, action years 2020–2023	315,581	—	Four action years; 2024 is read for outcomes only
Form R only	280,426	35,155	Form A short filings carry no release quantity
Manufacturing only (NAICS 31-33)	226,361	54,065	Study scope
Joined one-to-one to File 2A	226,361	0	Every Form R record has exactly one matching 2A record
After duplicate consolidation	224,185	2,176	Multiple forms per facility-chemical-year collapsed
Matched to the following year	205,447	18,738	The facility-chemical pair must recur to have an outcome
Eligible (positive base release, units agree)	163,501	41,946	Percentage change needs a non-zero denominator and a common unit
Development (2020–21, 2021–22, 2022–23)	123,535	—	Outcome computed
Reserved (2023–24)	39,966	—	Features only; outcome withheld

Note. The two analysis sets sum exactly to the eligible total: $123,535 + 39,966 = 163,501$. The largest single exclusion is the eligibility filter, not the matching step.

Table E6*Exploratory Data Analysis Insight Summary*

Figure or table	What it shows	Key insight	Why it matters	Decision or next step
Figure A1	Activity codes by W codes give way scheme and year	entirely to S codes after 2020; reported process-change prevalence rises across the same boundary	A behavioral trend and a coding change are confounded at the 2020–2021 boundary	Map all 70 codes to five types; assert zero unmapped; caveat the rise
Figure 1	Outcome and exposure balance	31.48% reduce; only 4.87% report any action	Outcome is balanced but exposure is rare	No resampling; exposure rarity limits action-type power
Figure A2	Release distributions	Mean 24,737 lb versus median 217 lb; skewness 34.6	Raw release cannot enter a model linearly	Use log release; check logit linearity against a flexible alternative in development diagnostics.
Figure A4	Production ratio and its missingness	Missingness is low; its mechanism cannot be confirmed from observed data	Missingness is low, but whether it is informative cannot be established from observed data	Median imputation plus a missingness indicator, learned within training data only; compare with complete-case results.
Figure 2	Reduction rate by transition	27.54%, 31.09%, 35.86%; non-overlapping intervals; $V = 0.073$	Year matters reliably but weakly; may partly reflect pandemic recovery and other temporal or reporting changes	Keep year as a control; repeat without 2020–2021
Figure 3	Rate and action prevalence by subsector	15.4-point spread; subsectors reporting most actions are not those reducing most	Industry confounds the raw comparison	Include the 15 qualifying subsectors plus an explicit Other Manufacturing category; use Chemical Manufacturing, NAICS 325, as the reference

Figure or table	What it shows	Key insight	Why it matters	Decision or next step
Figure 4	Action versus no action, and covariate balance	3.61-point raw gap, but 7 of 19 covariates imbalanced beyond 0.10	The raw gap is partly compositional	Adjusted model required before any claim
Figure A6	Rate by release decile	Current release is associated with the outcome, but the relationship is not strictly monotonic.	Starting release level and possible regression to the mean may confound the unadjusted association.	Retain log current release as a required control and assess functional form in the adjusted model.
Figure 5	Rate by production ratio	U-shaped; minimum 29.72% near a ratio of 1.00, rising to 36.19%	A linear term would miss the shape	Centered quadratic as provisional; test in full model
Table 1, Figure 6	Action types individually	Only material modification and operating practice survive Holm correction	Pooling action types would hide real differences	Report the five types separately
Figure A7	Hazard classification and action-year profiling	Dioxins reduce 6.7 points less often; carcinogen status shows no difference	Dioxins differ and use different units	Analyze dioxins separately as a sensitivity check
Figure A8	Facility clustering and 15-check quality scorecard (8 pass, 5 controlled, 2 limitations)	8.22 records per facility; 92.8% from multi-record facilities	Independence is violated	Facility-clustered standard errors in RQ2

Table E7*Feature Set and Planned Usage*

Feature	Original or engineered	Type	Reason for inclusion	Used in
log(1 + release_now)	Engineered	Continuous	Controls regression to the mean; RQ2, RQ4 raw scale is unusable at skewness 34.6	

Feature	Original or engineered	Type	Reason for inclusion	Used in
production_ratio_now (centered)	Original, centered	Continuous	Production ratio after training-fold median imputation, capping, and centering.	RQ2, RQ4
production_ratio_now squared	Engineered	Continuous	Captures the observed U-shape; provisional pending the full model	RQ2, possibly RQ4
ratio_missing	Engineered	Binary	Identifies records whose production ratio was originally missing after training-fold median imputation	RQ2, RQ4
subsector_now	Original	Categorical	Fifteen qualifying subsectors plus an explicit Other Manufacturing category; Chemical Manufacturing, NAICS 325, is the reference group.	RQ2, RQ4
span (action year)	Original	Categorical	Reduction rates rise across years; controls the pandemic-recovery trend in the explanatory model. Categorical span is not used as an RQ4 feature because the holdout contains an unseen action year.	RQ2 only
reported_any_action_now	Engineered	Binary	The primary RQ2 exposure	RQ2, RQ4
Five action-type indicators	Engineered	Binary	Action types behave differently; pooling would hide that	RQ3, RQ4
is_carcinogen_now, is_pbt_now	Original	Binary	Hazard controls specified in the analysis plan	RQ2, RQ4
is_dioxin_now, is_metal_now	Original	Binary	Chemical-class controls; dioxins also drive a separate sensitivity analysis	RQ4, sensitivity

Note. Identifier fields are excluded from every model by design and are retained only so that a flagged record can be traced back to a facility and chemical.

Table E8

Evaluation Metrics and Formulae

Metric	Formula	Interpretation	Used for
Reduction rate	$p = x / n$	Share of eligible records achieving at least a 10% reduction	RQ1, RQ3
Wilson score interval	$(p + z^2/2n \pm z \sqrt{[p(1-p)/n + z^2/4n^2]}) / (1 + z^2/n)$	Confidence interval for a proportion; reliable at extreme p	RQ1, RQ3
Pearson chi-square	$\chi^2 = \sum (O - E)^2 / E$	Tests whether rates differ across groups	RQ1, RQ3
Cramér's <i>V</i>	$V = \sqrt{(\chi^2 / [n \times \min(r-1, c-1)])}$	Effect size for a chi-square test; essential at large n	RQ1, RQ3
Rate difference	$d = p_1 - p_0$	Percentage-point gap against the no-action group	RQ3
Relative risk	$RR = p_1 / p_0$	How many times more likely a reduction is	RQ3
Katz interval for log RR	$SE(\ln RR) = \sqrt{(1/a - 1/n_1 + 1/c - 1/n_0)}$	Confidence interval for relative risk on the log scale	RQ3
Holm-adjusted p	Step-down over ordered <i>p</i> -values	Controls family-wise error across five comparisons	RQ3
Odds ratio	$OR = \exp(\beta)$	Adjusted association between action and reduction	RQ2
Clustered standard error	Sandwich estimator grouped by facility	Corrects for repeated records from the same facility	RQ2
Variance inflation factor	$VIF = 1 / (1 - R_j^2)$	Detects collinearity among predictors	RQ2, RQ4
Akaike information criterion	$AIC = 2k - 2 \ln(L)$	Compares non-nested functional forms	RQ2
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall share correctly classified	RQ4
Precision	$TP / (TP + FP)$	Of records flagged as at risk, the share truly at risk	RQ4
Recall (sensitivity)	$TP / (TP + FN)$	Of records truly at risk, the share the model flags	RQ4
F1 score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Balances precision and recall	RQ4
ROC AUC	Area under the true-positive versus false-positive curve	Discrimination; 0.50 is chance	RQ4
Brier score	$B = (1/n) \sum (\hat{p}_i - y_i)^2$	Mean squared error of predicted probabilities	RQ4

Note. In the RQ4 metrics the positive class is failure to achieve a reduction, because that is the class an EHS team

would act on. TP = true positives, TN = true negatives, FP = false positives, FN = false negatives.